



Number Series is a sequence formed out of numbers. In the number series questions, we will have to detect the rules/pattern/sequence that result in the formation of a number series.

Type of Number Series: According to recent exam, the number series categorized in three parts.

1. Missing number series
2. Wrong number series
3. Multiple number series

1. Missing number series: In this type of series and there is one number missing in the series and it is to be found by following the given pattern, in missing number series is a particular series in which a specific pattern is being followed by the numbers given, there are some different types of missing number series are given as follow

(a) Single stage difference: - In single stage difference, there is a continuous difference followed in the complete series

Example: 10, 15, 25, 40, 60, ? = 85

Here, pattern followed is + 5, +10, +15, +20, +25

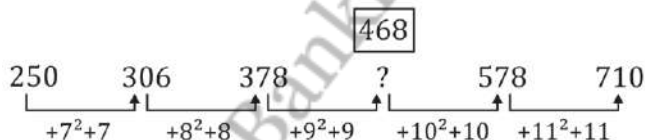
(b) Difference of Square or cube of numbers-based Series: - In such number series, the difference created is difference of squares of natural numbers

Example: 4, 8, 17, 33, 58, ? = 94

Here, the pattern of series is $+2^2$, $+3^2$, $+4^2$, $+5^2$, $+6^2$

In square number series, there is also a new pattern followed in the exams

Example:

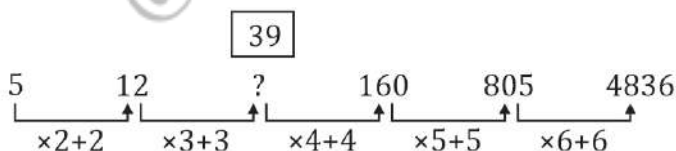


Here, the difference created is the square of the number and addition of that particular number

(c) Multiplication type series: - In such type of series, each term of the value is increased by a very large value

Example: 2, 4, 12, 48, ? = 240

Here, the pattern followed is $\times 2$, $\times 3$, $\times 4$, $\times 5$



Here, in this type of series the terms multiply and addition of same number

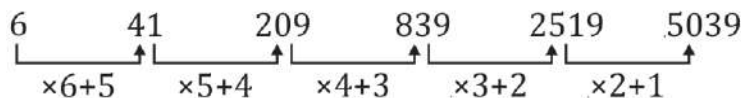
(d) Division type series: - In division type series, a particular number is being divided by a number and such pattern is being followed. In such series, the next number in the series is decreased highly from the previous number

Example - 3600, 1800, 600, 150, ? = 30

Here, the patter of series is $\div 2$, $\div 3$, $\div 4$, $\div 5$

(e) Mix Type Series: In such type of series, the pattern followed is based on more than one type/method. Such type of series is nowadays in trending in exams especially in mains exam

Example:



In such type of series, pattern followed is a number is first multiply/divide and then a same number or different number are added

- (f) **Prime number series:** In prime number series, the difference, addition, multiplication created in the series is consecutive prime numbers

Example – 12, 14, 17, 22, 29, ? = 40

Here, in this series the difference created is difference of prime number (consecutive) i.e. 2, 3, 5, 7, 11

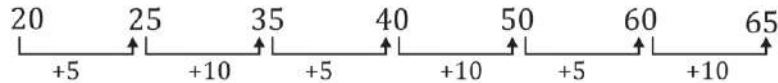
Wrong number series: In these types of series of numbers are given which follows some specific pattern, in which one of the numbers does not follow the given pattern of the series. This type of series is known as wrong number series. We can classify these series as in such type:

- (a) **Single stage difference wrong series:** In single stage difference wrong series, there is a continuous difference followed in the complete series and among them there is one wrong number in the whole series.

Example: 20, 25, 35, 40, 50, 60, 65

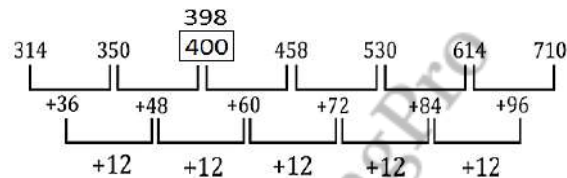
Here, the series follow the pattern +5, +10, +5, +10, +5, +10

So, wrong number = 60



- (b) **Double Stage difference:** In these types of series also value do not get increased or decreased by a very large value. In Double stage difference wrong number series, there is a specific pattern followed and among them there is a wrong number among the given series.

Example: wrong number = 398

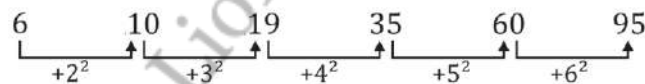


- (c) **Difference of Square or cube of numbers-based Series:** In these types of wrong number series, there is a specific pattern followed whose difference is squares and cubes of the number and among them there is a wrong number among them in the given series which does not follow the given pattern.

Example: 6, 10, 19, 35, 60, 95

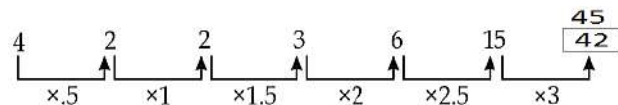
Here, the pattern of series is $+2^2$, $+3^2$, $+4^2$, $+5^2$, $+6^2$

So, wrong number = 95



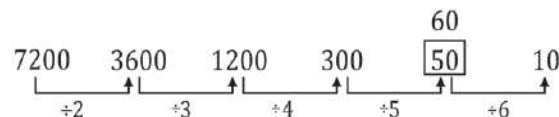
- (d) **Multiplication type series:** In this type, numbers of the series are the multiple of certain numbers which are based on the certain pattern.

Example:



- (e) **Division type series:** In division type series, terms of series is being divided by a number and a particular pattern is being followed. In such series, the next number in the series is decreased highly from the previous number.

Example:

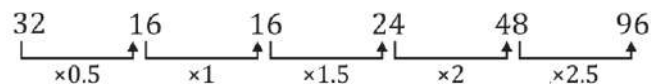


- (f) **Mix Type Series:** In mix type wrong number series, the difference obtained is followed by more than two pattern and among them a given number is wrong which does not follow the specified pattern

Example: 32, 16, 16, 24, 48, 96

Here, the pattern of series $\times 0.5$, $\times 1$, $\times 1.5$, $\times 2$, $\times 2.5$

So, wrong number = 96



1,	3,	9,	31,	129,	651
2,	—	—	—	?	

Similarly,

The diagram shows a horizontal line with arrows pointing right. Above the line are the numbers 2, 4, 11, 37, 153, and 771. Below the line, between each pair of consecutive numbers, is a label: $\times 1 + 2$, $\times 2 + 3$, $\times 3 + 4$, $\times 4 + 5$, and $\times 5 + 6$.

113,	170,	232,	303,	399,	556,	838.
93,	—	—	—	—	—	<u>m.</u>

113 170 232 303 399 556 838

+57 +62 +71 +96 +157 +282

+5 +9 +25 +61 +125

+4 +16 +36 +64

93 150 212 283 379 536 818

+57 +62 +71 +96 +157 +282

+5 +9 +25 +61 +125

+4 +16 +36 +64

- (i) To identify the multiplication pattern First start from right most side and check approx. multiple and then check from left most side for exact number.
- (ii) If numbers of a series decreases first and then increases continuously then first check product of $0.5 (\times 0.5 \mp n)$ or $(\times 1 - n)$.

1. 5 Questions of missing or wrong number series are asked in every bank exams either Pre or Mains. Except SBI PO mains and IBPS PO Mains. 2 to 3 questions of new pattern series asked in these both exams.
2. To solve the series we have to identify the pattern of series Usually the patterns based on
 - + Table
 - + Square
 - + Cube
 - + Square $\pm n$
 - + Cube $\pm n$
 - + Consecutive even or odd or prime no.
 - + Consecutive multiples.

- (a) Running Smoothly — Apply Difference
- (b) Bouncing Series — Check product
- (c) Zig-Zag — Add-Sub (Alternative)
Mul-Div (Alternative)
- (d) First Decrease then increase Continuously
— $\times 0.5 \pm n$
or
 $\times 1 - n$

**For More Study Material
Visit: adda247.com**

Missing Number Series

Directions (1-60): What will come in the place of question (?) mark in the following number series.

1. 900, 896, 912, 848, 1105, 80
(a) 900 (b) 912 (c) 896
(d) 1105 (e) 848
2. 7, 11, 20, 35, 61, 97
(a) 7 (b) 35 (c) 61
(d) 11 (e) 97
3. 2, 3, 7, 22, 89, 459
(a) 3 (b) 22 (c) 459
(d) 7 (e) 89
4. 2, 4, 12, 48, 240, 1448
(a) 4 (b) 12 (c) 240
(d) 1448 (e) 2
5. 2, 3.4, 6.2, 11.8, 25, 45.4, 90.2
(a) 25 (b) 11.8 (c) 45.4
(d) 3.4 (e) 90.2
6. 200, 193, 179, 158, ?, 95
(a) 135 (b) 133 (c) 132
(d) 130 (e) 128
7. 3, 43, 81, 115, 143, ?
(a) 163 (b) 172 (c) 166
(d) 160 (e) 168
8. 1, 6, 25, 76, 153, ?
(a) 152 (b) 154 (c) 153
(d) 155 (e) 156
39. 50, 54, 45, 61, 36, ?
(a) 66 (b) 72 (c) 75
(d) 80 (e) 84
10. 9, 45, 180, 540, ?, 1080
(a) 720 (b) 900 (c) 1080
(d) 1200 (e) 960
11. 23, 33, 18, 38, 13, ?
(a) 24 (b) 25 (c) 44
(d) 50 (e) 43
12. 5, 30, 150, 600, 1800, ?
(a) 3600 (b) 3300 (c) 3200
(d) 2000 (e) 2400
13. 196, 169, ?, 121, 100, 81
(a) 132 (b) 144 (c) 142
(d) 154 (e) 136
14. 10, 9, 16, 45, ?, 875
(a) 175 (b) 177 (c) 178
(d) 176 (e) 180
15. 7, 12, 19, 30, ?, 72
(a) 47 (b) 41 (c) 27
(d) 39 (e) 35
16. 44, 60, 80, 105, 136, ?
(a) 152 (b) 164 (c) 160
(d) 174 (e) 178
17. 10, 40, 140, 420, 1050, ?
(a) 2100 (b) 1400 (c) 1600
(d) 1900 (e) 2300
18. 32, 39, 34, 41, ?, 43
(a) 38 (b) 39 (c) 37
(d) 40 (e) 36
19. 1, 5, 14, 30, 55, ?
(a) 85 (b) 78 (c) 63
(d) 91 (e) 97
20. 5, 6, 13, 40, 161, ?
(a) 806 (b) 508 (c) 624
(d) 756 (e) 842
21. 1864, 1521, 1305, ?, 1116, 1089
(a) 1160 (b) 1180 (c) 1095
(d) 1205 (e) 1220
22. 18, ?, 9, 18, 72, 576
(a) 12 (b) 9 (c) 18
(d) 10 (e) 6
23. 12, 6.5, 7.5, 12.75, 27.5, ?
(a) 66.5 (b) 68.75 (c) 63.75
(d) 71.25 (e) None of these
24. 5, 15, 50, ?, 1030, 6185
(a) 210 (b) 205 (c) 225
(d) 200 (e) 195
25. 130, 154, 186, ?, 274, 330
(a) 216 (b) 220 (c) 240
(d) 226 (e) 230
26. 11, ?, 16, 21, 29, 41
(a) 12 (b) 14 (c) 15
(d) 13 (e) 11
27. 1800, ?, 60, 15, 5, 2.5
(a) 300 (b) 600 (c) 120
(d) 240 (e) 360
28. 4, 3, 4, 9, 32, ?
(a) 75 (b) 155 (c) 125
(d) 175 (e) 165
29. ?, 100, 150, 375, 1312.5
(a) 50 (b) 100 (c) 75
(d) 25 (e) 200
30. 0, 6, 24, 60, ?, 210
(a) 130 (b) 170 (c) 90
(d) 120 (e) 150
31. 200, 197, 185, 163, 130, ?
(a) 95 (b) 85 (c) 105
(d) 86 (e) 84

32. 15, 8, 9, 15, 32, ?
 (a) 98 (b) 66 (c) 80.5
 (d) 82.5 (e) 84.5
33. 5, 30, 150, 600, ?
 (a) 1200 (b) 1500 (c) 2400
 (d) 1800 (e) 600
34. 222, 110, 54, 26, ?
 (a) 10 (b) 12 (c) 8
 (d) 6 (e) 14
35. 104, ?, 96, 120, 88, 128
 (a) 112 (b) 96 (c) 116
 (d) 120 (e) 92
36. 12, 7, 8.5, 14.75, ?, 83
 (a) 30 (b) 24 (c) 32
 (d) 36 (e) 48
37. 12, 78, 395, 1584, 4755, ?
 (a) 9512 (b) 9516 (c) 9518
 (d) 9520 (e) 9580
38. 26, 53, 214, 1287, 10300, ?
 (a) 95000 (b) 100005 (c) 103000
 (d) 101005 (e) 103005
39. 4187, 2857, 2129, 1787, 1663, ?
 (a) 1647 (b) 1642 (c) 1627
 (d) 1637 (e) 1630
40. 27, 27, 54, 18, 72, ?
 (a) 19.6 (b) 16.8 (c) 18.8
 (d) 12.8 (e) 14.4
41. 15, 29, 56, 108, 208, 400, ?
 (a) 788 (b) 778 (c) 758
 (d) 764 (e) 768
42. 133, 183, 241, 307, 381, 463, ?
 (a) 557 (b) 553 (c) 559
 (d) 552 (e) 550
43. 1.21, 1.69, 2.89, 3.61, 5.29, ?
 (a) 84.1 (b) 4.41 (c) 6.25
 (d) 7.29 (e) 8.41
44. 36, 85, 121, 146, 162, 171, ?
 (a) 175 (b) 180 (c) 185
 (d) 160 (e) 165
45. 13, -21, 34, -55, 89, -144, ?
 (a) 235 (b) 236 (c) 237
 (d) 230 (e) 233
46. 60.75, ?, 27, 18, 12, 8
 (a) 45.45 (b) 48.25 (c) 40.5
 (d) 42 (e) 50.2
47. 8, 5, 7, 13.5, 31, ?
 (a) 82.5 (b) 67.25 (c) 72.25
 (d) 85.75 (e) 91.5
48. 209, ?, 217, 187, 255, 125
 (a) 203 (b) 207 (c) 196
 (d) 202 (e) 199
49. 3, 6, 12, 36, 180, ?
 (a) 1188 (b) 1264 (c) 1354
 (d) 1292 (e) 1332
50. 9, 28, 55, 66, 101, ?
 (a) 103 (b) 108 (c) 104
 (d) 111 (e) 124
51. 18, 62, 93, 113, ?, 128,
 (a) 120 (b) 122 (c) 123
 (d) 124 (e) 125
52. 100, 325, 69, 358, 34, ?
 (a) 395 (b) 245 (c) 365
 (d) 785 (e) 725
53. 10, 21, ?, 153, 450, 1341
 (a) 33 (b) 45 (c) 54
 (d) 36 (e) 39
54. ?, 12, 18, 27, 40.5, 60.75
 (a) 8 (b) 9 (c) 4
 (d) 3 (e) 2
55. 2, 6, 21, 110, ?, 8558
 (a) 687 (b) 577 (c) 547
 (d) 677 (e) 777
56. ?, 192, 1536, 6144, 12288, 12288
 (a) 12 (b) 10 (c) 6
 (d) 15 (e) 36
57. 1726, 996, 506, 208, 54, ?
 (a) -4 (b) 2 (c) -2
 (d) 0 (e) 18
58. 1296, ?, 648, 324, 486, 405
 (a) 6 (b) 4 (c) 0
 (d) 5 (e) 2
59. 8, ?, 16, 44, 158, 715
 (a) 2 (b) 0 (c) 16
 (d) 4 (e) 8
60. ?, 70, 346, 1378, 4126, 8242
 (a) 12 (b) 6 (c) 8
 (d) 36 (e) 24

Wrong Number Series

Directions (1-30): Find the wrong number in the following number series:

1. 4, 5.1, 7.3, 10.6, 15, 20, 27.1

- (a) 5.1 (b) 4 (c) 7.3
 (d) 20 (e) 27.1

2. 2, 3, 8, 31, 154, 924, 6460

- (a) 924 (b) 6460 (c) 154
 (d) 8 (e) 31

3. 251, 252, 254, 227, 243, 118, 154
(a) 251 (b) 252 (c) 227
(d) 243 (e) 154
4. 141, 156, 147, 162, 153, 165, 159
(a) 156 (b) 153 (c) 147
(d) 165 (e) 159
5. 2, 6, 10, 19, 36, 69, 134
(a) 134 (b) 69 (c) 6
(d) 2 (e) 10
6. 0.5, 2, 1, 4, 32, 512, 16384
(a) 1 (b) 2 (c) 4
(d) 32 (e) 512
7. 1, 3, 9, 31, 128, 651, 3913
(a) 9 (b) 1 (c) 128
(d) 31 (e) 3913
8. 291, 147, 75, 39, 22, 12, 7.5
(a) 22 (b) 291 (c) 147
(d) 75 (e) 7.5
9. 26, 27, 34, 58, 106, 186, 306
(a) 26 (b) 34 (c) 58
(d) 106 (e) 27
10. 5.9, 6, 6.1, 6.4, 7.9, 18.5, 112.9
(a) 6 (b) 5.9 (c) 6.1
(d) 18.5 (e) 112.9
11. 330, 80, 280, 120, 250, 130, 240
(a) 330 (b) 130 (c) 280
(d) 240 (e) 80
12. 2, 3, 6, 15, 45, 156.5, 630
(a) 2 (b) 15 (c) 3
(d) 156.5 (e) 630
13. 36, 20, 12, 8, 6, 5.5, 4.5
(a) 8 (b) 36 (c) 5.5
(d) 4.5 (e) 6
14. 1, 3, 9, 31, 128, 651, 3913
(a) 31 (b) 3 (c) 1
(d) 3913 (e) 128
15. 2, 3, 10, 40, 172, 885, 5346
(a) 40 (b) 885 (c) 172
(d) 3 (e) 10
16. 5, 8, 16, 26, 50, 98, 194
(a) 5 (b) 194 (c) 8
(d) 16 (e) 98
17. 18, 36, 12, 24, 8, 18
(a) 18 (b) 36 (c) 8
(d) 24 (e) 12
18. 120, 149, 166, 185, 208, 237
(a) 185 (b) 166 (c) 149
(d) 120 (e) 208
19. 204, 210, 235, 251, 376, 412
(a) 204 (b) 210 (c) 235
(d) 251 (e) 412
20. 79, 54, 75, 58, 71, 62, 65
(a) 62 (b) 65 (c) 71
(d) 58 (e) 75
21. 50, 53, 67, 100, 160, 255, 395
(a) 100 (b) 160 (c) 255
(d) 395 (e) 50
22. 1600, 1604, 1544, 1760, 1248, 2248
(a) 1600 (b) 1544 (c) 1760
(d) 2248 (e) 1604
23. 275, 75, 165, 33, 99, 19.8
(a) 99 (b) 165 (c) 75
(d) 19.8 (e) 275
24. 20000, 500, 25, 1.5, 0.5, 0.2
(a) 1.5 (b) 500 (c) 25
(d) 0.5 (e) 0.2
25. 1001, 934, 863, 790, 711, 630, 539
(a) 539 (b) 790 (c) 1001
(d) 630 (e) 790
26. 2.2, 30, 6.6, 16.5, 49.5, 173.25, 693
(a) 2.2 (b) 30 (c) 16.5
(d) 49.5 (e) 693
27. 82.2, 5.5, 71.5, 6.5, 58.5, $\frac{58.5}{7}$
(a) 6.5 (b) 71.5 (c) 5.5
(d) $\frac{58.5}{7}$ (e) 82.2
28. 143, 305, 440, 548, 629, 685, 710
(a) 143 (b) 710 (c) 548
(d) 685 (e) 440
29. 22.4, 80, 128, 166.4, 195.2, 214.4, 224
(a) Series is correct (b) 22.4
(c) 224 (d) 166.4
(e) 80
30. 35, 86, 168, 290, 460, 686, 970
(a) Series has two wrong number
(b) Series is correct
(c) 686
(d) 970
(e) 168
- Direction (31-36):** There is a wrong number in each of the given number series, first you have to find the wrong number and then find the correct number from the option which replace that wrong number in these series.
31. 12, 19, 27, 90, 114, 329, 671
(a) 367 (b) 673 (c) 375
(d) 377 (e) 373

32. 36, 28, 38, 26, 40, 56, 42
(a) 26 (b) 22 (c) 24
(d) 48 (e) 36
33. 124, 264, 316, 348, 364, 372, 376
(a) 248 (b) 252 (c) 256
(d) 254 (e) 258
34. 178, 207, 230, 249, 266, 289, 290
(a) 269 (b) 277 (c) 279
(d) 281 (e) 283
35. 14, 8, 10, 24, 104, 842, 13600
(a) 858 (b) 846 (c) 848
(d) 838 (e) 836
36. 7, 19, 34, 45, 59, 69, 81
(a) 86 (b) 72 (c) 84
(d) 82 (e) 88

Direction (37-60): Find the wrong number in the following number series given below.

37. 6000, 3000, 2000, 1500, 1875, 1000, $857\frac{1}{7}$
(a) 3000 (b) 1500 (c) 1875
(d) 1000 (e) $857\frac{1}{7}$
38. 86.4, 21.6, 108, 27, 136, 33.75, 168.75
(a) 86.4 (b) 136 (c) 108
(d) 27 (e) 33.75
39. 175, 188, 208, 236, 331, 447, 784
(a) 784 (b) 331 (c) 236
(d) 175 (e) 188
40. 8403, 7693, 7125, 6699, 6415, 6273, 6272
(a) 7693 (b) 7125 (c) 6699
(d) 6272 (e) 6273
41. 1, 6.5, 18.5, 61, 249.5, 1254, 7523.5
(a) 1 (b) 6.5 (c) 18.5
(d) 61 (e) 1254
42. 554, 545, 520, 471, 350, 182
(a) 554 (b) 545 (c) 520
(d) 182 (e) 471
43. 24, 48, 88, 170, 332, 654, 1296
(a) 48 (b) 24 (c) 88
(d) 170 (e) 1296
44. 1, 24, 288, 1729, 5184, 7776, 5832
(a) 1 (b) 24 (c) 288
(d) 1729 (e) 5184

45. 1250, 367, 204, 161, 142, 129, 116
(a) 1250 (b) 116 (c) 367
(d) 161 (e) 129
46. 577, 596, 624, 688, 835, 1122, 1636
(a) 596 (b) 1122 (c) 1636
(d) 577 (e) 835
47. 1071, 1068, 1077, 1101, 1161, 1281, 1491
(a) 1071 (b) 1161 (c) 1077
(d) 1068 (e) 1491
48. 17, 36, 30, 180, 172, 1720, 1708
(a) 17 (b) 36 (c) 30
(d) 180 (e) 1708
49. 36, 86, 168, 290, 460, 684, 976
(a) 86 (b) 36 (c) 684
(d) 976 (e) 168
50. 5934, 894, 174, 54, 30, 24, 20
(a) 174 (b) 54 (c) 30
(d) 24 (e) 20
51. 179, 180, 187, 210, 257, 339, 469
(a) 180 (b) 179 (c) 210
(d) 339 (e) 469
52. 1200, 1000, 1080, 864, 604.8, 362.88, 181.44
(a) 1200 (b) 1000 (c) 1080
(d) 181.44 (e) 604.8
53. 214, 428, 424, 848, 844, 1688, 422
(a) 214 (b) 428 (c) 422
(d) 848 (e) 1688
54. 789, 796, 814, 853, 923, 1030, 1196
(a) 789 (b) 853 (c) 1030
(d) 923 (e) 1196
55. 640, 129, 516, 103, 412, 82.2, 328.8
(a) 129 (b) 412 (c) 82.2
(d) 328.8 (e) 640
56. 131, 194, 244, 279, 305, 320, 310
(a) 310 (b) 131 (c) 194
(d) 244 (e) 279
57. 18, 28, 41, 57, 75, 94, 117
(a) 57 (b) 18 (c) 28
(d) 117 (e) 75

Advance Number Series

Directions (1-3): Find the pattern of the series and answer the question given below.

Series: X, 168, 172, 145, Y, Z

1. Find the 8th term of the series.

- (a) 271 (b) -324 (c) -271

- (d) 324 (e) -218

2. Which of the following statement/s is are correct.

- (i) X is greater than Y
(ii) Z is square of even number
(iii) X is less than Z

- (a) All (i), (ii) & (iii) (b) Both (ii) & (iii)
 (c) Both (i) & (iii) (d) Both (i) & (ii)
 (e) None of the above

3. Find the Sum of the Y and Z.

- (a) 182 (b) 167 (c) 197
 (d) 198 (e) 209

Directions (4-6): Find the pattern of the series and answer the question given below.

Series I: 8, 9, 17, 52, $25A+7$, 1036, B

4. Find which of the statement/s is/are correct.

- (i) $B/113 + A > 67$
 (ii) $\sqrt{B} + 26 = 79$
 (iii) A is completely divisible by first term of the series I.

- (a) All (i), (ii) & (iii) (b) Both (i) & (iii)
 (c) Only (iii) (d) Both (i) & (ii)
 (e) None of the above

5. Find the value of B.

- (a) 6615 (b) 6516 (c) 6115
 (d) 5215 (e) 6215

6. $A+5$ is the first term of the series II, which follows the same pattern as series I, then find the fourth term of the series II.

- (a) 27 (b) 82 (c) 327
 (d) 1636 (e) 89

Direction (7-10): Solve the given series and answer the following questions.

Series: $a + 15$, $a + 16.2$, $a + 18.6$, ____, ____, M

Note: (i) x and y are prime number less than 10. ($y > x$)

- (ii) $xy = a$
 (iii) $2a = 10y$.

7. Find which of the series follows the same pattern as above series.

- I. 25.8, 27, 29.4, 33, 37.8, 43.8
 II. 112, 113.2, 115.6, 119.2, 124, 130
 III. 325, 326.2, 328.6, 332.2, 337, 343
 (a) Both I & II (b) Both I & III (c) All I, II & III
 (d) None of the above (e) Only I

8. Find the seventh term of the given series.

- (a) 75.2 (b) 68 (c) 83.6
 (d) 77.2 (e) 92.8

9. Find the value of M.

- (a) $9y + x$ (b) $12x + 8$
 (c) $2a - 2$ (d) All (a), (b) & (c)
 (e) None of the above

10. If 7 is added to the value of $2a$, then the resultant is?

- (a) Divisible by 7 (b) Multiple of 11
 (c) Prime number (d) (a) & (b)
 (e) None of the these

Directions (11-12): Find the pattern of the given series and answer the following question given below.

Series: ?, 8, 72, 4608, 2880000

11. If the ratio of X to the missing term of the above series is 5:2, respectively, then find the value of $3X$.

- (a) 12 (b) 30 (c) 21
 (d) 9 (e) 15

12. If X is 140% more than the missing term of the above series, then find the value of X.

- (a) 7.4 (b) 2.6 (c) 4.4
 (d) 9.6 (e) 6.6

Direction (13-14): Solve the number series to answer the following questions.

$\sqrt[n]{A}$ 29 B C 437 B + 687 $\frac{1294}{\sqrt[n]{A}}$

Note:

- (i) 'n' is a positive integer.
 (ii) Difference between 29 and B is X, which has 3 factors excluding X itself, those are 13, 5 and A but when we include X then number of factors become 4.

13. What is the value of $A + C$?

- (a) 221 (b) 222 (c) 223
 (d) 224 (e) 225

14. What is the LCM of $B + 20$ and A?

- (a) 101 (b) 102 (c) 114
 (d) 104 (e) 115

Direction (15-17): Solve the series to answer the following questions.

Series I: 60 120 24 48 9.6 19.2

Series II: 100 W X Y Z 32

Note: Both series follows the same patterns

15. Find the difference between $(Y+X)$ and $(Y+Z)$

- (a) 12 (b) 24 (c) 14
 (d) 4 (e) 25

16. If a new series starts with the value of $(W-X)$ following the same logic mentioned in the above series, then what is the 3rd term of the newly formed series?

- (a) 120 (b) 121 (c) 144
 (d) 64 (e) 25

17. If roots of quadratic equation are $\frac{Y}{W}$ and $(\frac{W-X}{16})$, then find the quadratic equation whose variable is r?

- (a) $r^2 - \frac{90r}{5} + \frac{7}{11} = 0$ (b) $r^2 - \frac{32r}{5} + \frac{8}{25} = 0$
 (c) $r^2 - \frac{87r}{4} + 3 = 0$ (d) $r^2 - \frac{52r}{5} + 4 = 0$
 (e) None of the above

Direction (18-19): Solve the series and answer the following questions given below.

Series 1: 87 B $(2a+15)$ $(30^2 + 6)$ $(103 + Z)$ 1076 1101

Series 2: $(a+58)$ 492 529 Y 639 712 749

Note – $B = 87 + a$

- 18.** Find the value of a.
 (a) 361 (b) 342 (c) 356
 (d) 344 (e) 352
- 19.** Find the value of Z.
 (a) 654 (b) 924 (c) 656
 (d) 644 (e) 652
- 20.** Find the value of B
 (a) 548 (b) 428 (c) 568
 (d) 448 (e) 528
- 21.** If x and y are the terms of series 1, then find the sum of x and y.
 Series 1 – 6, 9, 14, 21, x, 45, y
 (a) 94 (b) 92 (c) 83
 (d) 91 (e) 97

Direction (22-23): One number is wrong, and one number is missing in the given number series. Solve the number series and answer the questions given below.

(I) 0.5, 5, 60, 480, 2880, 11520, P

- 22.** Find the wrong number of the series.
 (a) 5 (b) 60 (c) 480
 (d) 2880 (e) 0.5
- 23.** If the wrong number of the series is A, then find which of the statement/s is/are correct.
 (I) $P - 2880 = 4032A$
 (II) P is completely divisible by A
 (III) $P > 4320A$
 (a) Both (I) and (II) (b) Only (I)
 (c) Only (III) (d) All (I), (II) & (III)
 (e) None of these

Directions (24-25): Find the pattern of the series and answer the questions given below.

Series I. a, d, c, 669, 833, 1017, 1221

Series II. b, b+18, 336, 354, 372, 390

Series III. 456, 472, 408, 552, z

- 24.** Which of the following option is correct.
 (a) $a+d - c > z$
 (b) $a+d - c < z$
 (c) $a+d = b + z + 102$
 (d) (a) and (c) both are correct.
 (e) (b) and (c) both are correct.
- 25.** Which of the following statement/s is/are incorrect.
 (A) d is a prime number.
 (B) a and z both are an odd number.
 (C) c is divided by 7, then the quotient is $\frac{1}{4}$ of b.
 (a) Only (B) (b) Both (B) and (C)
 (c) Both (C) & (A) (d) Only (C)
 (e) Both (A) & (B)

Direction (26-29): There are two series I and II given below. Both series containing wrong term and both the series follow different patterns. Solve the series and answer the following questions given below.

Series I: a, b+5, c+8, 87, 412, 2185, 13326

Series II: b+4, c+8, 36, a+50, 79, 111, 152

Note:

- (i) $10bx^2 - (21 - a)x + 3 = 0$, roots of the equation $z/10$ and $y/20$ (where $z/10 > y/20$).
 (ii) any multiple of y give a number with unit digit zero.
 (iii) $c - z = 9$ and LCM of z and 5 is 15.

- 26.** Find the LCM of y, z and c.
 (a) 60 (b) 10 (c) 15
 (d) 25 (e) 20
- 27.** Find the difference between wrong term of series II and correct term of series I in place of wrong term.
 (a) Odd number (b) Prime number
 (c) Composite number (d) Even number
 (e) Both (d) and (c)
- 28.** If wrong term of series I is subtracted from 25, then find the resultant term.
 (a) 60 (b) 10 (c) 15
 (d) 25 (e) 18
- 29.** If a new series starting from z follows same logic as series I, then find the third term.
 (a) 6 (b) 16 (c) 15
 (d) 25 (e) 20
- 30.** In which of the following series value of X is not satisfied properly?
Series I: 7 14 X 37 53 72
Series II: 48 X 26 53 213 1705
Series III: 8 15 X 35 48 63
 Value of X is 24.
 (a) only I and III (b) only III (c) only I
 (d) only II (e) None of these

Direction (31 - 32): Solve the given series and answer the question given below.

18 10 12 27 X P

- 31.** If ' $Z + 4 = X - 27$ ', then which of the following is true about value of 'Z'?
 I. Z is a perfect square
 II. Z is an odd number
 III. Z is an even number
 (a) only I and II (b) only III (c) only I
 (d) only II (e) None of these
- 32.** Find the value of P?
 (a) 112 (b) 924 (c) 926
 (d) 954 (e) 901

Directions (33-35): There is a wrong number in these series. Find the wrong number & pattern of the given series and answer the questions given below.

Series A: 24, 31.5, 46.5, 69, 98, 136.5, 181.5

Series B: 5926, 886, 166, 46, 22, 18, 14

Series C: 11, 18, 44, 107, 231, 445, 788

33. X, Y & Z are the wrong number of the series A, B & C respectively. Find the relation between X, Y & Z.
 (a) $X > Y < Z$ (b) $X > Y > Z$ (c) $X \leq Y < Z$
 (d) $X > Y \geq Z$ (e) $X = Y < Z$
34. If series D is following the pattern of series C. If first term of series D is 23 and P & Q being the third & sixth term respectively of series D, then find the difference between P & Q.
 (a) 540 (b) 195 (c) 298
 (d) 402 (e) 338
35. P & Q are the correct terms of series A & series B respectively. If R is equal to the square of the larger root of the equation $x^2 - 15x = -9^2 + 5^2$, then which of the following statement/s is/are correct?
 (i) $Q + R = P$
 (ii) $\frac{P}{3} + 15 = R - Q$
 (iii) $\sqrt{R} + P = Q \times 7 - 5$
 (a) Only (i) & (ii) (b) Only (ii) & (iii) (c) Only (i)
 (d) Only (ii) (e) Only (iii)
36. If X is the correct number which replace the wrong number of the series 229, 256, 320, 441, 661, 1004, then which of following statement/s is or are correct.
 I. X is completely divisible by 15.
 II. $(X+2)$ is a prime number.
 (a) Both I and II correct
 (b) Both I and II incorrect
 (c) I correct and II incorrect
 (d) I incorrect and II correct
 (e) Can't determine
37. Series (A): 1332, 612, 492, Y, 462, 460
 If $\frac{Y-18}{5}$ is a fourth term of series (B) and follow same pattern as of the series (A), then find the first term of series (B).
 (a) 954 (b) 882 (c) 1045
 (d) 1198 (e) 892
38. 49, ?, 82, 145, 269, 484, 826
 If $? = A + 6$, then find the nearest square of A.
 (a) 4 (b) 8 (c) 5
 (d) 11 (e) 7
39. Subsequent terms in all the given three number series should follow the pattern "1 more than the square of consecutive odd numbers".
 A. 32, 82, 164, 286, 456, 682
 B. 140, 166, 216, 338, 508, 798
 C. 400, 410, 435, 485, 567, 689
 Which of the following series does not follow the pattern?
 (a) Only C (b) Only A (c) Both A and C
 (d) Both B and C (e) None of these

Direction (40 – 41): Given below three series I, II & III and each series has a wrong number. The number that should come in place of wrong number in series I, II & III is a, b & c respectively.

I. 5, 6, 8, 14, 38, 168, 878, 5918

II. 32, 544, 593, 809, 832, 898, 907

III. 18, 38, 124, 500, 2504, 15028, 105200

40. If $m^2 = \sqrt{a + 98}$, then which of the following statement/s is or are true.

(A) $(\frac{b}{m} + 0.5) = \text{resultant is an integer.}$

(B) $10m = c$

(C) $m + \frac{(b-42)}{18} \geq c$

(a) Only (A) and (C) (b) Only (B) and (C)

(c) Only (A) and (B) (d) All three

(e) Only C

41. If y is two times of the highest root of equation $n^2 - cn + (2.5a - 11)$, then which of the following option/s is true.

(a) $(y - c) = 2^3$ (b) Value of $\frac{b}{y}$ is more than 16.

(c) Value of $\frac{a}{y}$ is an integer

(d) None of these

(e) Both (a) and (b)

Direction (42-44): A set of 2 number series is given below. Both series follow a similar pattern. Answer the questions based on it using information given below:

I. 5, 6, 13, 40, 161, 806

II. 2, p, q, r, s

42. Find the value of s.

(a) 22 (b) 82 (c) 89

(d) 84 (e) 77

43. If nth term of series II is 2677, then find the value of 'n'.

(a) 11 (b) 13 (c) 9

(d) 7 (e) 8

44. Find the value of $(r - pq)$.

(a) -3 (b) 5 (c) 0

(d) -1 (e) 1

Direction (45-46): Given below are two number series. Series I is a missing series while series II is a wrong number series which follows pattern of series I only.

I. 11, P, 181, 350, 639, 1000

2. 242, 251, 255, 280, 329, 450, 619

45. If 2 is added in successive terms of a series starting with the nearest prime of P, then find the 5th term of this new series

1. 67 2. 65 3. 69

4. 63

(a) either 1 or 3 (b) either 2 or 3 (c) either 2 or 4

(d) either 3 or 4 (e) either 1 or 2

46. If 'y' is the wrong number of the series II, then find the value of '2y+1'.

- (a) 493 (b) 561 (c) 503
(d) 511 (e) 901

Direction (47 – 48): Calculate the value of (a) and (b) and answer the questions given below:

(a), (b), 102, 104, 107, 113, 128, 173

47. From (b) a new series started. The second term of series is $(b + 2^2)$ and the third term of series is $(b + 3^3)$, fourth term of series is $(b + 4^4)$. Find the 5th term of series.

- (a) $98 + 5^5$ (b) $99 + 5^5$ (c) $100 + 5^5$
(d) $101 + 5^5$ (e) None of these

48. If double the value of (a) is X, then find the difference between X and 4th term of series given.

- (a) 82 (b) 84 (c) 86
(d) 88 (e) 80

49. 'X' is the nth term of the given series and 'Y' is the $(n + 1)$ th term of the given series. Choose the correct statement from the following statements (i), (ii) & (iii).

701, 349, 173, 85, 41, 19, 8, 2.5

- (i) $X = 2Y + 3$
(ii) $Y = 3X + 2$
(iii) Difference between 1st and 2nd term is twice of the difference between 2nd and 3rd term and so on.
(a) Only (i) and (iii) (b) Only (i) and (ii)
(c) Only (ii) and (iii) (d) Only (i)
(e) All (i), (ii) & (iii)

50. In the following questions, a series is given in which one term is wrong with which another series started. You have to find the wrong term in the given series then starting from that, find IVth term of new series so formed.

1, 3, 6, 21, 88, 445, 2676

- (a) 39 (b) 37 (c) 25
(d) 33 (e) 35

Direction (51 – 52): Series I: 8, 9, 15, 25, 42, 68, 105

Series II: 60, 180, 450, 900, 1350, 1380, 675

51. Wrong term of series I is the nearest square of which of the given term?

- (a) 5 (b) 4 (c) 8
(d) 7 (e) 3

52. If wrong term in series II is 'N', then which statement is true about $(\frac{N}{30} + 1)$?

- (i) It's a prime number.
(ii) Sum of the digits is less than 9.
(iii) It's a nearest multiple of 5 and its remainder is 4.
(a) Only (ii) and (iii) follow
(b) Only (i) and (ii) follow
(c) Only (i) follow

(d) Non follow

(e) All (i), (ii) and (iii) follow

Direction (53 – 54): Read the data carefully and answer the following the questions.

20, (a), (b), (c), (d), 65

$$a - 20 = X^2 + Y$$

$$b - a = (X + 1)^2 + Y(b > a)$$

$$c - b = (X + 2)^2 + Y(c > b)$$

$$d - c = (X + 3)^2 + Y(d > c)$$

Given, X = HCF of M & N, which are prime numbers

And, Y = Smallest root of $z^2 - z - 6 = 0$

53. How many numbers are divisible by 3?

- (a) a & c Only (b) a, b & d Only (c) b & d Only
(d) c & d Only (e) a & d Only

54. Find the value of 'd'?

- (a) 19 (b) 21 (c) 28
(d) 42 (e) 48

55. Given below is the series in which one number is wrong. Consider this wrong number as the value of A in the second series and find the value of C based on the pattern of first series

1, 3, 6, 21, 88, 445, 2676

(A), (B), (C)

- (a) 685 (b) 136 (c) 33
(d) 10 (e) 30

Directions (57-60):- Given questions are based on a missing series pattern and following that pattern find the relation between P, Q and R.

57. 300, 324, 384, 504, P, 1050

450, 474, Q, 654, 864, 1200

200, 224, 284, 404, R, 850

- (a) $P > Q < R$ (b) $P < Q < R$ (c) $P = Q > R$
(d) $P < Q = R$ (e) $P < Q > R$

58. 2700, 5400, P, 7200, 1440, 8640

2100, Q, 1400, 5600, 1120, 6720

1500, 3000, 1000, R, 800, 4800

- (a) $P > Q > R$ (b) $P < Q = R$ (c) $P < Q > R$
(d) $P = Q > R$ (e) $P < Q < R$

59. 35, 70, 210, P, 4200, 25200

140, 280, Q, 3360, 16800, 100800

40, 80, 240, R, 4800, 28800

- (a) $P > Q < R$ (b) $P < Q > R$ (c) $P < Q < R$
(d) $P = Q < R$ (e) $P > Q > R$

60. 7, 15, 47, 191, Q, 5754

9, 19, 59, P, 1199, 7199

11, 23, 71, 287, R, 8639

- (a) $P > Q < R$ (b) $P < Q > R$ (c) $P < Q < R$
(d) $P = Q < R$ (e) $P > Q > R$

Directions (61-62): There are three series given in a question, you have to find value of x, y and z in each question and then establish relation among them.

61. 1, 601, 721, 751, (x), 766
753, 769, (y), 765, 763, 764
23, 24, 50, 153, (z), 3085
(a) $x = y = z$ (b) $x > y > z$ (c) $x > y = z$
(d) $x = y > z$ (e) $x = z > y$

62. 102, 107, 117, 134, (x)
130, 115, 135, 110, (y)
(z), 80, 120, 300, 1050
(a) $x > y = z$ (b) $y < x = z$ (c) $x = y = z$
(d) $y > x > z$ (e) $x = z < y$

Directions (64-66): Given below is the sequence of series. Analyze the pattern of the series and answer the given following questions.

64. 1, 3, 9, 31, 129, 651
2, —, —, —, —, ?
(a) 625 (b) 37 (c) 153
(d) 771 (e) 631

65. 4, 2, 2, 3, 6, 15, 45
If $(2835)^n$ is the term of the sequence where n is the nth term of sequence then find 'n'.
(a) 8th (b) 9th (c) 10th
(d) 11th (e) 12th

66. A series is 113, 170, 232, 303, 399, 556, 838. Another series is 93, —, —, —, —, m. Which follows same pattern as given number series. Then m = ?
(a) 808 (b) 443 (c) 626
(d) 818 (e) 909

Directions (67-69): There is a wrong term given in each series I, II and III given below. Find the wrong number and pattern of the given series and answer the questions given below.

Series I: 48, 60, 150, 750, 5625, 150000

Series II: 1500, 1000, 720, 288, 57.6, 0

Series III: 15, 23, 86, 110, 325, 372

67. If P, Q and R are the correct terms which come in the place of wrong term of series I, II and III respectively, then find the relation among P, Q and R.

- (a) $P > Q < 2R$ (b) $\frac{P}{5} > 1.5Q > R$
(c) $P = (6Q + 300) > 20R$ (d) $P > 6Q = 15R$
(e) None of these

68. X and Z is the two wrong terms of series I & III respectively, while Y is the correct terms which come in the place wrong term of series II. The second term of a newly formed series IV is equal to $(X - 5Y + Z)$. If series IV follow the pattern same as of series III, then find difference between first and fourth term of series

IV.

- (a) 93 (b) 95 (c) 73
(d) 81 (e) 99

69. X, Y and Z are the three wrong terms of series I, II and III respectively. Find the value of A and B in the given number series below and also find which of the given statement/s is or are true.

Series: 6, A, 14, 38, 158, 878, B

(i) $X + Z + 10 = B + 17^2 - 2 \times (10)^2$

(ii) $Y = A \times 5^3$

(iii) $\sqrt[3]{A} + Z = \frac{Y}{2.5}$

- (a) Both (ii) and (iii) (b) Both (i) and (ii)
(c) Only (i) (d) Only (ii)
(e) All (i), (ii) and (iii)

Direction (70-71): There are two series A and B given below. Solve both series and answer the questions given below.

Series A: 17, z, $(9y - 35)$, $(48^2 + 44)$, $(50^2 + a)$, 2998, 3167

Series B: $(x+36)$, 132, y, $(b^2 - 3)$, 327, 374, $(21^2 + 7)$, 495

Note: (i) z = Twice of 7th term of series B + $(9^2 + 1)$

(ii): a and b are positive integers

70. If $P = (a + y)$, then Which of the following statement/s is or are true.

I. $P = 8x + (b+7)$

II. $P = \frac{49}{2} \times b + \sqrt{529}$

III. $2P = z - 148$

- (a) Both I and II (b) Both I and III (c) Only I
(d) All I, II and III (e) Both II and III

71. $2^p \times 7^q = bx$, then find the value of $(\frac{z}{2} - 2a)^{p-q}$.

- (a) 4489 (b) 4761 (c) 5329
(d) 3721 (e) 5041

72. There are two series I and II given below. Solve both series and answer the questions given below.

Series I: 76, X, 556, 636, 676, 696, Y

Series II: A, 148, 177, 223, 286, B, 463

Which of the following statement/s is or are incorrect.

P. $Y = 5A + \sqrt{676}$

Q. Difference between X and B is an even number

R. $2A + Y = 2B + (X - 12^2)$

- (a) Both P and Q (b) Only P (c) Only Q
(d) All P, Q and R (e) Only R

Direction (73 - 74): There are three series I, II and III are given. Each series has a wrong number. The number that should come in place of wrong number in series I, II and III are x, y & z respectively.

I. 8, 10, 20, 30, 96, 122, 360, 390

II. 16, 10, 18, 11, 19.5, 11.5, 20.625

III. 26, 28, 112, 118, 944, 9440, 11448

73. If $a^2 = \sqrt{(x+2)} + z$, then which of the following statement/s is or are true.

- (i) $2a \leq (y + \frac{1}{4})$ (ii) $2^a > y$
 (iii) $x + 60a < z$
 (a) Both (i) and (ii) (b) All (i), (ii) and (iii)
 (c) Only (iii) (d) Both (i) and (iii)
 (e) Only (i)

74. If 'b' is three times of the difference between highest and lowest root of equation $p^2 - 4yp + \frac{3}{2}x$, then which of the following option/s is true.

- (a) $2(x - \frac{z}{3}) - 5 = b$
 (b) Value of $\frac{z}{b}$ is more than 18.
 (c) Value of '4yb' is an integer
 (d) None of these
 (e) All (a), (b) and (c)

Direction (75 – 76): Calculate the value of (x) and (y) in the given number series and answer the questions given below:

2, 1730, 2594, 3890, 7130, (x), (y)

75. From (y) a new series started and the second term of the new series will be $(y \div 5)$, the third term of new series will be $(y \div 10)$, fourth term of series is $(y \div 20)$, then which option will be true about 5th term of the new formed series.

- (a) 5th term is an integer number
 (b) 5th term < 1800.5
 (c) 5th term is a whole number
 (d) 5th term $> (12)^3$
 (e) Both (b) and (d)

76. If $A = (\frac{x}{50} - 45.4)$, then find for the value (approx.) $(y \div A)$ which of the following option will be definitely true.

- I. $(y \div A) < 6^3$ II. $(y \div A) > 14^2$ III. $(y \div A)^2 > y$
 (a) Only I (b) Only III (c) Only II
 (d) Both II and III (e) Both I and II

Direction (77–79): There are three number series given below i.e. I, II, & III where the missing number in these series are X, Y & Z respectively. Find the value of X, Y & Z in the following number series and answer the questions given below.

I. 119, 125, 135, 153, X, 229, 295

II. 4, 28, 168, 840, 3360, Y, 20160

III. 105, 111, Z, 119, 89, 131, 75

77. Find the relation between X, Y & Z.

- (a) $X < Y < Z$ (b) $Z > X \leq Y$ (c) $Y > X > Z$
 (d) $Z \geq Y = X$ (e) $X = Y < Z$

78. Find which among the given statement (s) is/are true.

- (i) HCF of X and Z is $\sqrt[3]{27}$.
 (ii) ' $\frac{Y}{Z-9} + 32$ ' is a perfect square.
 (iii) Difference between Y and X is divisible by 4.
 (a) Only (ii) (b) Only (i) & (ii) (c) None of these
 (d) Only (iii) (e) Only (i) & (iii)

79. If Pattern of a series IV is same as pattern of the series III and first term of the series IV is X, then find the fifth term of the series IV.

- (a) 197 (b) 189 (c) 167
 (d) 209 (e) 153

80. Given below three series I, II & III and each series has a wrong number. The number that should come in place of wrong number in series I, II & III is a, b & c respectively.

I. 5, 6, 8, 14, 38, 168, 878, 5918

II. 32, 544, 593, 809, 832, 898, 907

III. 18, 38, 124, 500, 2504, 15028, 105200

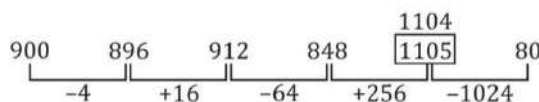
81. If $P = (2c+a+b)$, then which of the following term square is nearest to P.

- (a) 32 (b) 33 (c) 29
 (d) 30 (e) 34

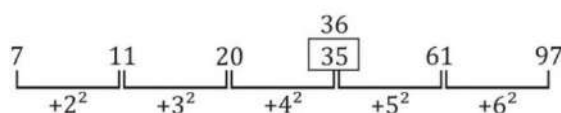
Solution

Missing Number Series

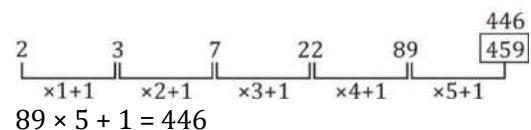
1. (d):



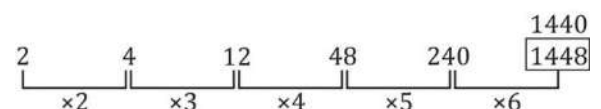
2. (b):



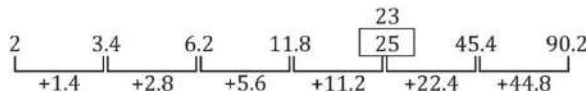
3. (c):



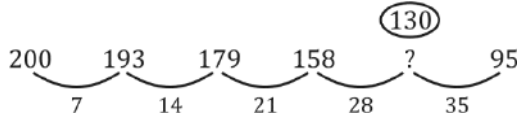
4. (d):



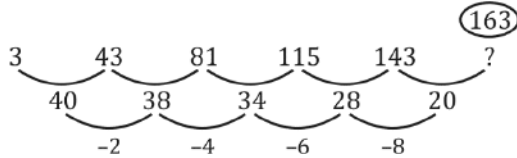
5. (a):



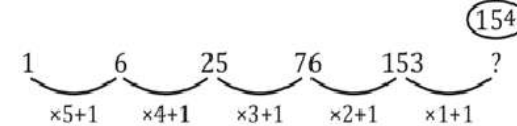
6. (d):



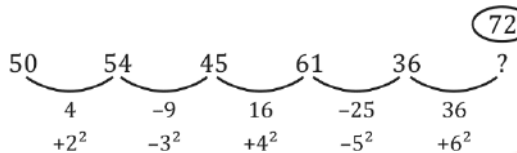
7. (a):



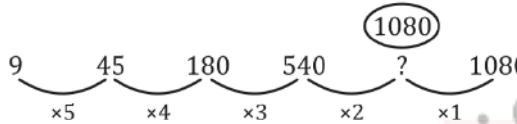
8. (b):



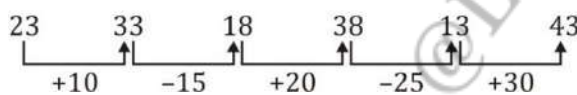
9. (b):



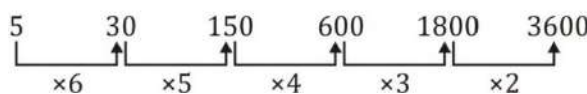
10. (c):



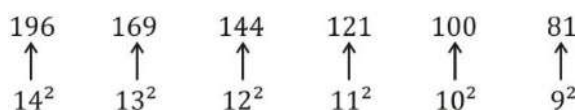
11. (e)



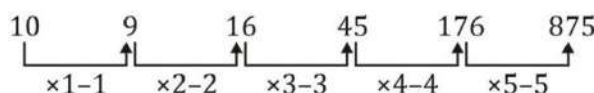
12. (a):



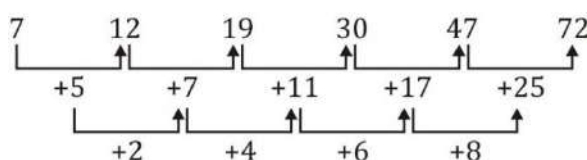
13. (b):



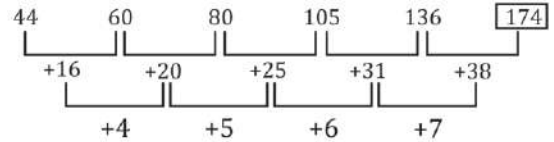
14. (d):



15. (a):

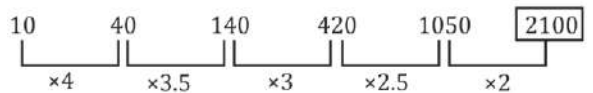


16. (d): Pattern of series -



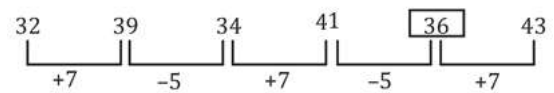
So, missing number is 174.

17. (a): Pattern of series -



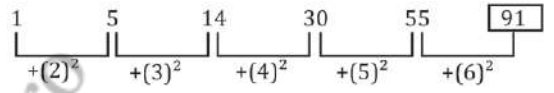
So, missing number is 2100.

18. (e): Pattern of series -



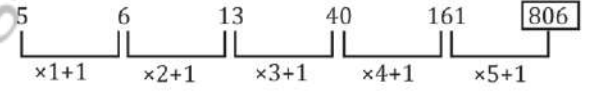
So, missing number is 36.

19. (d): Pattern of series -



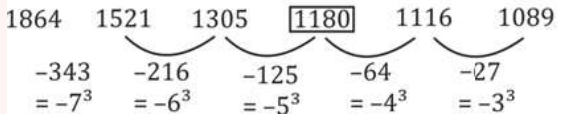
So, missing number is 91.

20. (a): Pattern of series -



So, missing number is 806.

21. (b):



22. (b): 18, ?, 9, 18, 72, 576

$$\begin{aligned} 18 \times 0.5 &= 9 \\ 9 \times 1 &= 9 \\ 9 \times 2 &= 18 \\ 18 \times 4 &= 72 \\ 72 \times 8 &= 576 \end{aligned}$$

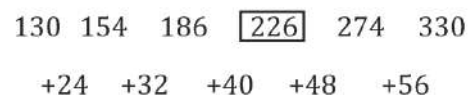
23. (d): $12 \times 0.5 + 0.5 = 6.5$

$$\begin{aligned} 6.5 \times 1 + 1 &= 7.5 \\ 7.5 \times 1.5 + 1.5 &= 12.75 \\ 12.75 \times 2 + 2 &= 27.5 \\ 27.5 \times 2.5 + 2.5 &= 71.25 \end{aligned}$$

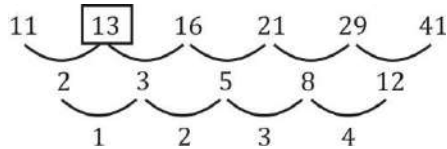
24. (b): $5 \times 2 + 5 = 15$

$$\begin{aligned} 15 \times 3 + 5 &= 50 \\ 50 \times 4 + 5 &= 205 \\ 205 \times 5 + 5 &= 1030 \\ 1030 \times 6 + 5 &= 6185 \end{aligned}$$

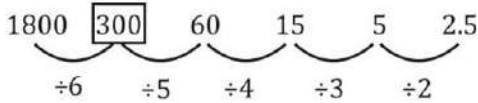
25. (d):



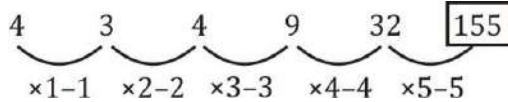
26. (d):



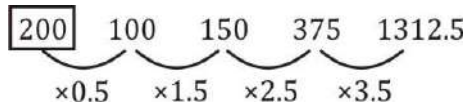
27. (a):



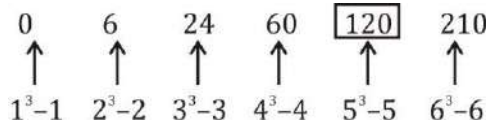
28. (b):



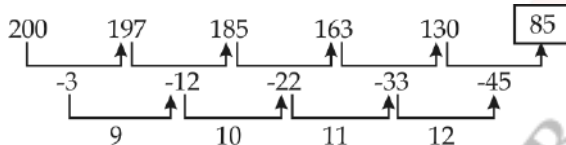
29. (e):



30. (d):



31. (b):



32. (d): The pattern is $-15 \times 0.5 + 0.5 = 8$

$$8 \times 1 + 1 = 9$$

$$9 \times 1.5 + 1.5 = 15$$

$$15 \times 2 + 2 = 32$$

$$32 \times 2.5 + 2.5 = 82.5$$

33. (d): The pattern is $\times 6, \times 5, \times 4, \times 3, \times 2, \dots$

$$\therefore ? = 600 \times 3 = 1800$$

34. (b): The pattern is: $\times 0.5 - 1, \times 0.5 - 1, \times 0.5 - 1, \times 0.5 - 1, \dots$

$$? = 26 \times 0.5 - 1 = 12$$

35. (a): The pattern is: $+ 8, - 16, + 24, - 32, + 40$

$$? = 104 + 8 = 112$$

36. (c): Pattern of series

$$12 \times 0.5 + 1 = 7$$

$$7 \times 1 + 1.5 = 8.5$$

$$8.5 \times 1.5 + 2 = 14.75$$

$$? = 14.75 \times 2 + 2.5 = 32$$

$$32 \times 2.5 + 3 = 83$$

37. (a): Pattern of series —

$$12 \times 6 + 6 = 78$$

$$78 \times 5 + 5 = 395$$

$$395 \times 4 + 4 = 1584$$

$$1584 \times 3 + 3 = 4755$$

$$? = 4755 \times 2 + 2 = 9512$$

38. (e): $26 \times 2 + 1 = 53$

$$53 \times 4 + 2 = 214$$

$$214 \times 6 + 3 = 1287$$

$$1287 \times 8 + 4 = 10300$$

$$? = 10300 \times 10 + 5 = 103005$$

39. (d): Pattern of series —

$$4187 - (11^3 - 1) = 2857$$

$$2857 - (9^3 - 1) = 2129$$

$$2129 - (7^3 - 1) = 1787$$

$$1787 - (5^3 - 1) = 1663$$

$$? = 1663 - (3^3 - 1) = 1637$$

40. (e): Pattern of series

$$27 \div 1 = 27$$

$$27 \times 2 = 54$$

$$54 \div 3 = 18$$

$$18 \times 4 = 72$$

$$? = 72 \div 5 = 14.4$$

41. (e): $15 \times 2 - 1 = 29$

$$29 \times 2 - 2 = 56$$

$$56 \times 2 - 4 = 108$$

$$108 \times 2 - 8 = 208$$

$$208 \times 2 - 16 = 400$$

$$400 \times 2 - 32 = 768$$

42. (b): 133 183 241 307 381 463 553

$$50 \ 58 \ 66 \ 74 \ 82 \ 90$$

$$8 \ 8 \ 8 \ 8 \ 8$$

43. (e): $1.1^2 = 1.21$

$$1.3^2 = 1.69$$

$$1.7^2 = 2.89$$

$$1.9^2 = 3.61$$

$$2.3^2 = 5.29$$

$$2.9^2 = 8.41$$

44. (a): 36 85 121 146 162 171 175

$$49 \ 36 \ 25 \ 16 \ 9 \ 4$$

45. (e): $13 - (-21) = 34$

$$-21 - (-34) = -55$$

$$34 - (-55) = 89$$

$$-55 - (89) = -144$$

$$89 - (-144) = 233$$

46. (c): Pattern of the given series:

$$60.75 \times \frac{2}{3} = 40.5$$

$$40.5 \times \frac{2}{3} = 27$$

$$27 \times \frac{2}{3} = 18$$

$$18 \times \frac{2}{3} = 12$$

$$12 \times \frac{2}{3} = 8$$

47. (a): The pattern of the series is:

$$8 \times 0.5 + 1 = 5$$

$$5 \times 1 + 2 = 7$$

$$7 \times 1.5 + 3 = 13.5$$

$$13.5 \times 2 + 4 = 31$$

$$31 \times 2.5 + 5 = 82.5$$

48. (b): The pattern of the series is:

$$209 - (1^3 + 1) = 207$$

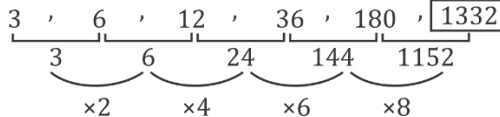
$$207 + (2^3 + 2) = 217$$

$$217 - (3^3 + 3) = 187$$

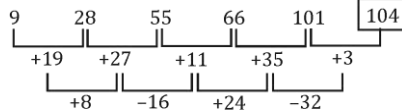
$$187 + (4^3 + 4) = 255$$

$$255 - (5^3 + 5) = 125$$

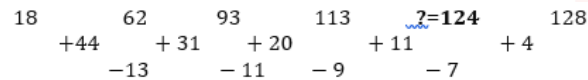
49. (e): The pattern of the series is:



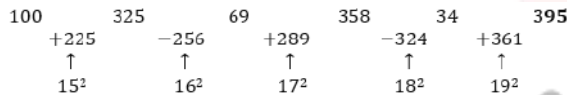
50. (c): The pattern of the series is:



51. (d): Pattern of the series -



52. (a): Pattern of the series -



53. (c): Pattern of the series -

$$10 \times 3 - 9 = 21$$

$$21 \times 3 - 9 = 54$$

$$54 \times 3 - 9 = 153$$

$$153 \times 3 - 9 = 450$$

$$450 \times 3 - 9 = 1341$$

54. (a): Pattern of the series -

$$8 \times \frac{3}{2} = 12$$

$$12 \times \frac{3}{2} = 18$$

$$18 \times \frac{3}{2} = 27$$

$$27 \times \frac{3}{2} = 40.5$$

$$40.5 \times \frac{3}{2} = 60.75$$

55. (e): Pattern of the series -

$$2 \times 2 + 2 = 6$$

$$6 \times 3 + 3 = 21$$

$$21 \times 5 + 5 = 110$$

$$110 \times 7 + 7 = 777$$

$$777 \times 11 + 11 = 8558$$

56. (a): Pattern of series -

$$12 \times 16 = 192$$

$$192 \times 8 = 1536$$

$$1536 \times 4 = 6144$$

$$6144 \times 2 = 12288$$

$$12288 \times 1 = 12288$$

57. (a): Pattern of series -

$$(12^3 - 2) = 1726$$

$$(10^3 - 4) = 996$$

$$(8^3 - 6) = 506$$

$$(6^3 - 8) = 208$$

$$(4^3 - 10) = 54$$

$$(2^3 - 12) = -4$$

58. (c): Pattern of series -

$$1296 - 1296 = 0$$

$$0 + 648 = 648$$

$$648 - 324 = 324$$

$$324 + 162 = 486$$

$$486 - 81 = 405$$

59. (e): Pattern of series -

$$8 \times 0.5 + 4 = 8$$

$$8 \times 1.5 + 4 = 16$$

$$16 \times 2.5 + 4 = 44$$

$$44 \times 3.5 + 4 = 158$$

$$158 \times 4.5 + 4 = 715$$

60. (a): Pattern of series -

$$12 \times 6 - 2 = 70$$

$$70 \times 5 - 4 = 346$$

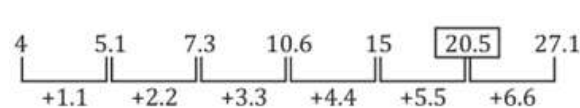
$$346 \times 4 - 6 = 1378$$

$$1378 \times 3 - 8 = 4126$$

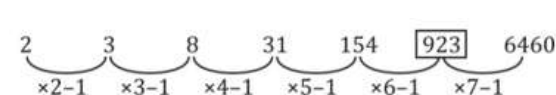
$$4126 \times 2 - 10 = 8242$$

Wrong Number Series

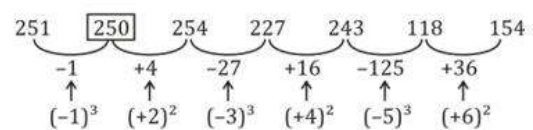
1. (d):



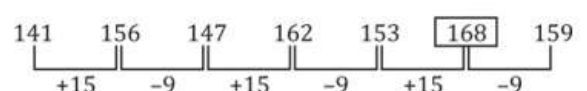
2. (a):



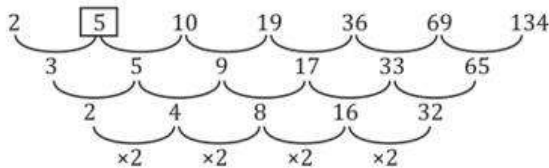
3. (b):



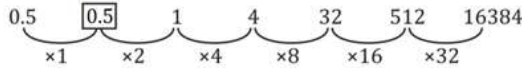
4. (d):



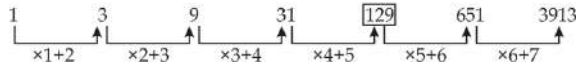
5. (c):



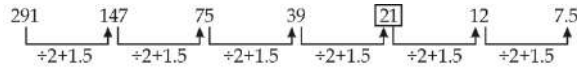
6. (b):



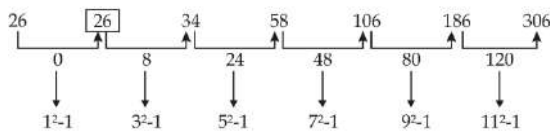
7. (c):



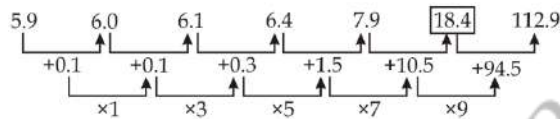
8. (a):



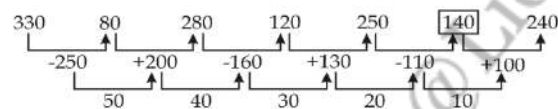
9. (e)



10. (d):

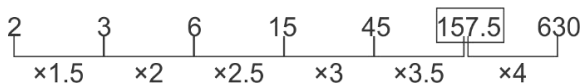


11. (b):



12. (d): Wrong number = 156.5

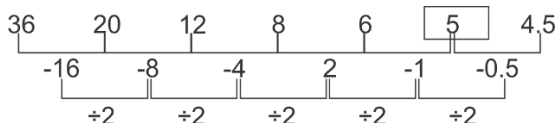
Pattern of series-



So, there should be 157.5 in place of 156.5.

13. (c): Wrong number = 5.5

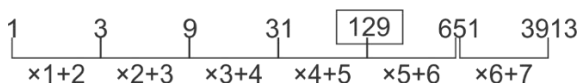
Pattern of series-



So, there should be 5 in place of 5.5

14. (e): Wrong number = 128

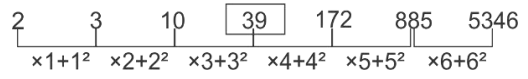
Pattern of series -



So, there should be 129 in place of 128

15. (a): Wrong number = 40

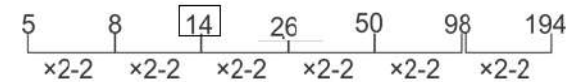
Pattern of series -



So, there should be 39 in place of 40.

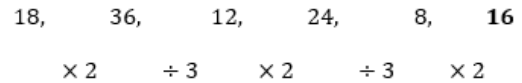
16. (d): Wrong number = 16

Pattern of series -

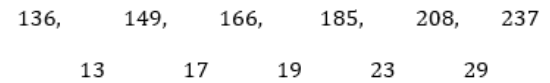


So, there should be 14 in place of 16.

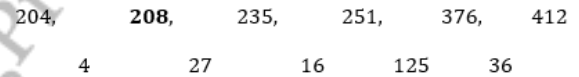
17. (a): The pattern of the series:



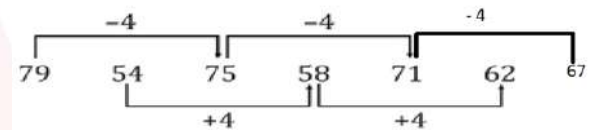
18. (d): The pattern of the series:



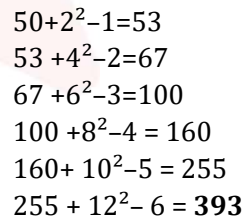
19. (b): The pattern of the series:



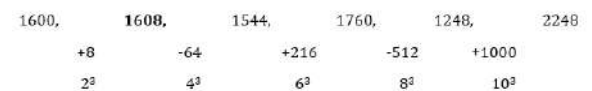
20. (b): Pattern of the series:



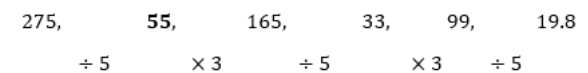
21. (d): Pattern of the series:



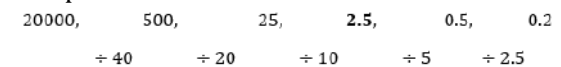
22. (e): The pattern of the series:



23. (c): The pattern of the series:

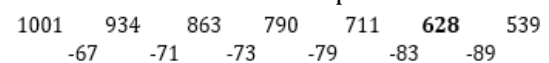


24. (a): The pattern of the series:



25. (d): The pattern of the series:

Subtraction of consecutive prime number



26. (b): Pattern of the series:

$$2.2 \times 1.5 = 3.3$$

$$3.3 \times 2 = 6.6$$

$$6.6 \times 2.5 = 16.5$$

$$16.5 \times 3 = 49.5$$

$$49.5 \times 3.5 = 173.25$$

$$173.25 \times 4 = 693$$

$$\text{Wrong number} = 30$$

27. (e): Pattern of the series:

$$82.5 \times \frac{1}{15} = 5.5$$

$$5.5 \times 13 = 71.5$$

$$71.5 \times \frac{1}{11} = 6.5$$

$$6.5 \times 9 = 58.5$$

$$58.5 \times \frac{1}{7} = \frac{58.5}{7}$$

$$\text{Wrong number} = 82.2$$

28. (d): Pattern of series –

$$143 + (27 \times 6) = 305$$

$$305 + (27 \times 5) = 440$$

$$440 + (27 \times 4) = 548$$

$$548 + (27 \times 3) = 629$$

$$629 + (27 \times 2) = 683$$

$$683 + (27 \times 1) = 710$$

29. (a): Pattern of series –

$$22.4 + (9.6 \times 6) = 80$$

$$80 + (9.6 \times 5) = 128$$

$$128 + (9.6 \times 4) = 166.4$$

$$166.4 + (9.6 \times 3) = 195.2$$

$$195.2 + (9.6 \times 2) = 214.4$$

$$214.4 + (9.6 \times 1) = 224$$

All numbers are correct.

30. (a): Pattern of series –

$$36 + (7^2 + 1) = 86$$

$$86 + (9^2 + 1) = 168$$

$$168 + (11^2 + 1) = 290$$

$$290 + (13^2 + 1) = 460$$

$$460 + (15^2 + 1) = 686$$

$$686 + (17^2 + 1) = 976$$

31. (d): Wrong number = 671

Pattern of series –

$$12 + (2^3 - 1) = 19$$

$$19 + (3^2 - 1) = 27$$

$$27 + (4^3 - 1) = 90$$

$$90 + (5^2 - 1) = 114$$

$$114 + (6^3 - 1) = 329$$

$$329 + (7^2 - 1) = 377$$

So, 377 will come in the place of 671

32. (c): Wrong number = 56

Pattern of series –

$$36 - 8 = 28$$

$$28 + 10 = 38$$

$$38 - 12 = 26$$

$$26 + 14 = 40$$

$$40 - 16 = 24$$

$$24 + 18 = 42$$

So, 24 will in the place of 56

33. (b): Wrong number = 264

Pattern of series –

$$124 + 2^7 = 252$$

$$252 + 2^6 = 316$$

$$316 + 2^5 = 348$$

$$348 + 2^4 = 364$$

$$364 + 2^3 = 372$$

$$372 + 2^2 = 376$$

So, 252 will come in the place of 264

34. (c): Wrong number = 289

Pattern of series –

$$178 + 29 = 207$$

$$207 + 23 = 230$$

$$230 + 19 = 249$$

$$249 + 17 = 266$$

$$266 + 13 = 279$$

$$279 + 11 = 290$$

So, 279 will come in the place of 289

35. (c): Wrong number = 842

Pattern of series –

$$14 \times 0.5 + 1 = 8$$

$$8 \times 1 + 2 = 10$$

$$10 \times 2 + 4 = 24$$

$$24 \times 4 + 8 = 104$$

$$104 \times 8 + 16 = 848$$

$$848 \times 16 + 32 = 13600$$

36. (d): Wrong number = 81

Pattern of series –

7,	19,	34,	45,	59,	69,	82
↓	↓	↓	↓	↓	↓	↓
+12	+15	+11	+14	+10	+13	
↓	↓	↓	↓	↓	↓	
+3	-4	+3	-4	+3		

So, 82 will come in the place of 81

37. (c): The pattern of the series:

6000,	3000,	2000	1500,	1200,	1000,	$\frac{6000}{7}$
$\times \frac{1}{2}$	$\times \frac{2}{3}$	$\times \frac{3}{4}$	$\times \frac{4}{5}$	$\times \frac{5}{6}$	$\times \frac{6}{7}$	

Wrong number = 1875

38. (b): Wrong number = 136

Pattern of series –

$$86.4 \div 4 = 21.6$$

$$21.6 \times 5 = 108$$

$$108 \div 4 = 27$$

$$27 \times 5 = 135$$

$$135 \div 4 = 33.75$$

$$33.75 \times 5 = 168.75$$

39. (e): The pattern of the series:

175,	187,	208,	236,	331,	447,	784
12	21	28	95	116	337	
9	7	67	21	221		
2^3+1	3^2-2	4^3+3	5^2-4	6^3+5		

Wrong number = 188

40. (d): The pattern of the series:

8403,	7693,	7125,	6699,	6415,	6273,	6273
710	568	426	284	142	0	
(71×10)	(71×8)	(71×6)	(71×4)	(71×2)	(71×0)	

Wrong number = 6272

41. (e): Wrong no. = 1254

Pattern of series –

$$1 \times 1 + 5.5 = 6.5$$

$$6.5 \times 2 + 5.5 = 18.5$$

$$18.5 \times 3 + 5.5 = 61$$

$$61 \times 4 + 5.5 = 249.5$$

$$249.5 \times 5 + 5.5 = 1253$$

$$1253 \times 6 + 5.5 = 7523.5$$

42. (d): Wrong no. = 182

Pattern of series –

$$554 - 9 = 545$$

$$545 - 25 = 520$$

$$520 - 49 = 471$$

$$471 - 121 = 350$$

$$350 - 169 = 181$$

43. (a): Wrong no. = 48

Pattern of series –

$$24 \times 2 - 2 = 46$$

$$46 \times 2 - 4 = 88$$

$$88 \times 2 - 6 = 170$$

$$170 \times 2 - 8 = 332$$

$$332 \times 2 - 10 = 654$$

$$654 \times 2 - 12 = 1296$$

44. (d): Wrong no. = 1729

Pattern of series –

$$1 \times 24 = 24$$

$$24 \times 12 = 288$$

$$288 \times 6 = 1728$$

$$1728 \times 3 = 5184$$

$$5184 \times 1.5 = 7776$$

$$7776 \times 0.75 = 5832$$

45. (b): Wrong no. = 116

1250	367	204,	161,	142,	129,	118
883	163	43	19	13	11	
720	120	24	6	2		
6!	5!	4!	3!	2!		

46. (e): Wrong no. = 835

577	596,	624,	688,	833,	1122,	1636
19	28	64	145	289	514	
9	36	81	144	225		
3^2	6^2	9^2	12^2	15^2		

47. (d): Wrong no. = 1068

1071	1071,	1077,	1101,	1161,	1281,	1491
$(1^3 - 1)$	$(2^3 - 2)$	$(3^3 - 3)$	$(4^3 - 4)$	$(5^3 - 5)$	$(6^3 - 6)$	

48. (b): Wrong no. = 36

17	34,	30,	180,	172,	1720,	1708
$\times 2$	-4	$\times 6$	-8	$\times 10$	-12	

49. (c): Wrong number = 684

Pattern of series –

$$36 + (7^2 + 1) = 86$$

$$86 + (9^2 + 1) = 168$$

$$168 + (11^2 + 1) = 290$$

$$290 + (13^2 + 1) = 460$$

$$460 + (15^2 + 1) = 686$$

$$686 + (17^2 + 1) = 976$$

So, 686 should be in the place of 684.

50. (e): Wrong number = 20

Pattern of the series:

5934,	894,	174,	54,	30,	24,	22
5040	720	120	24	6	2	
7!	6!	5!	4!	3!	2!	

51. (a): Wrong number = 180

Pattern of the series:

179,	179,	187,	210,	257,	339,	469
0	8	23	47	82	130	
8	15	24	35	48		
$(3^2 - 1)$	$(4^2 - 1)$	$(5^2 - 1)$	$(6^2 - 1)$	$(7^2 - 1)$		

52. (b): Wrong number = 1000

Pattern of the series:

1200,	1200,	1080,	864,	604.8,	362.88,	181.44
$\times \frac{100}{100}$	$\times \frac{90}{100}$	$\times \frac{80}{100}$	$\times \frac{70}{100}$	$\times \frac{60}{100}$	$\times \frac{50}{100}$	

53. (c): Wrong number = 422

Pattern of the series:

214,	428,	424,	848,	844,	1688,	1684
$\times 2$	-4	$\times 2$	-4	$\times 2$	-4	

54. (c): Wrong number = 1030

Pattern of the series:

789,	796,	814,	853,	923,	1034,	1196
7	18	39	70	111	162	
11	21	31	41	51		

55. (e): Wrong number = 640

Pattern of the series:

646,	129,	516,	103,	412,	82.2,	328.8
$-1 + 5$	$\times 4$	$-1 + 5$	$\times 4$	$-1 + 5$	$\times 4$	

56. (a): Wrong number = 310

Pattern of the series:

131, 194, 244, 279, 305, 320, 330
 8^2-1 7^2+1 6^2-1 5^2+1 4^2-1 3^2+1

57. (a): Wrong no. = 57

Pattern of series -

$$4^2 + 2 = 18$$

$$5^2 + 3 = 28$$

$$6^2 + 5 = 41$$

$$7^2 + 7 = 56$$

$$8^2 + 11 = 75$$

$$9^2 + 13 = 94$$

$$10^2 + 17 = 117$$

Advance Number Series

Sol. (1-3):

The pattern of the series:

X, 168, 172, 145, Y, Z
 -1^3 $+2^2$ -3^3 $+4^2$ -5^3

$$X = 168 + 1 = 169$$

$$Y = 145 + 16 = 161$$

$$Z = 161 - 125 = 36$$

1. (c): 8th term of the series = $36 + 6^2 - 7^3$
 $= 36 + 36 - 343 = -271$

2. (d): From (i) X is greater than Y (It is correct)
 From (ii) Z is square of even number (It is correct)
 From (iii) X is less than Z (It is incorrect)

3. (c): Required sum = $161 + 36 = 197$
 (4-6)

The pattern of the series I:

8, 9, 17, 52, 25A+7, 1036, B
 $\times 1 + 1$ $\times 2 - 1$ $\times 3 + 1$ $\times 4 - 1$ $\times 5 + 1$ $\times 6 - 1$

$$52 \times 4 - 1 = 25A + 7$$

$$200 = 25A$$

$$A = 8$$

$$B = 1036 \times 6 - 1 = 6215$$

4. (a): From (i) $B/113 + A > 67$
 $6215/113 + 8 > 67$
 $63 > 67$
 It is correct
 From (ii) $\sqrt{B + 26} = 79$
 $\sqrt{6215 + 26} = 79$
 $79 = 79$
 It is correct
 From (iii) $A = 8$
 First term of the series I = 8
 A is completely divisible by first term of the series I.
 It is correct

5. (e): $B = 6215$

6. (b): $A = 8 + 5 = 13$
 13 = First term of the series II
 The pattern of the series II:

13, 14, 27, 82
 $\times 1 + 1$ $\times 2 - 1$ $\times 3 + 1$

Sol. (7-10):

Prime number less than 10 be 2, 3, 5 & 7.

Let $x = 2$ and $y = 3$

Then $a = xy = 6$

$2a = 12$ and $10y = 30$ (which is not equal)

Let $x = 3$ and $y = 5$

Then $a = xy = 15$

$2a = 30$ and $10y = 50$ (which is not equal)

Let $x = 5$ and $y = 7$

Then $a = xy = 35$

$2a = 70$ and $10y = 70$ (which is equal)

So, $x = 5$, $y = 7$ & $a = 35$

The pattern of the series:

50, 51.2, 53.6, 57.2, 62, 68
 $+1.2$ $+2.4$ $+3.6$ $+4.8$ $+6$

7. (c):

I. 25.8, 27, 29.4, 33, 37.8, 43.8

$+1.2$ $+2.4$ $+3.6$ $+4.8$ $+6$

II. 112, 113.2, 115.6, 119.2, 124, 130

$+1.2$ $+2.4$ $+3.6$ $+4.8$ $+6$

III. 325, 326.2, 328.6, 332.2, 337, 343

$+1.2$ $+2.4$ $+3.6$ $+4.8$ $+6$

All the series follows the given pattern.

8. (a): Seventh term of the given series = $68 + 7.2 = 75.2$

9. (d): (a) $9y + x = 63 + 5 = 68$

(b) $12x + 8 = 60 + 8 = 68$

(c) $2a - 2 = 70 - 2 = 68$

All (a), (b) & (c)

10. (d): $2a + 7 = 70 + 7$
 $= 77$

(11-12): Pattern of the series:

? = 4, 8, 72, 4608, 2880000

$\times 2$ $\times 9$ $\times 64$ $\times 625$

2^1 3^2 4^3 5^4

11. (b): ATQ, $\frac{x}{4} = \frac{5}{2} \Rightarrow x = 10$

Required value = $3x = 30$

12. (d): Required value = $\frac{240}{100} \times 4 = 9.6$

Sol. (13 - 14): $B - 29 = X$

Factors of $X = 13, 5, A$

$A = 1$

$X = 65$

$B = 94$

So, series is: 1, 29, 94, C, 437, 781, 1294

And the pattern follows;

1 29 94 220 437 781 1294
 $3^3 + 1$ $4^3 + 1$ $5^3 + 1$ $6^3 + 1$ $7^3 + 1$ $8^3 + 1$

$C = 94 + 125 + 1 = 220$

13. (a): $A + C = 221$

14. (c): LCM of (1, 114) = 114

Sol. (15 - 17):

Series I: 60 120 24 48 9.6 19.2

Follows pattern = $\times 2, \div 5, \times 2, \div 5, \dots$

60 120 24 48 9.6 19.2
 $\times 2$ $\div 5$ $\times 2$ $\div 5$ $\times 2$

Series II: 100 W X Y Z 32

$W = 200$

$X = 40$

$Y = 80$

$Z = 16$

15. (b): Required answer = $120 - 96 = 24$

16. (d): First term = 160

Second term = 320

Third = 64

17. (d): Root I: 0.4 root II: 10

Required quadratic = $\left(r - \frac{4}{10}\right)(r - 10) = 0$

$5r^2 - 52r + 20 = 0$

$r^2 - \frac{52r}{5} + 4 = 0$

Sol (18 - 20):

Solving series 2:

$(a+58)$ 492 529 Y 639 712 749

37 73 37

So, the series follows the pattern of alternate addition of 73 and 37 respectively

$492 - a - 58 = 73$

$361 = a$

$Y = 529 + 73 = 602$

$B = 448$

Solving series 1-

87 448 $(2a+15)=737$ 906 $(103 + Z)$ 1076 1101

361 289 169

Series 1 follows the addition of square of prime number stating from 19.

$906 + 121 = 103 + Z$

$Z = 924$

18. (a): $361 = a$

19. (b): $Z = 924$

20. (d): $B = 448$

21. (a): $6+3=9$

$9+5=14$

$14+7=21$

$21+11=32$

$32+13=45$

$45+17=62$

Required answer = $x+y=32+62=94$

Sol. (22 -23):

0.5, 6, 60, 480, 2880, 11520, **P= 23040**
 $\times 12$ $\times 10$ $\times 8$ $\times 6$ $\times 4$ $\times 2$

Wrong number = 5

22. (a): Wrong number = 5

23. (d): $A=5$

(I) $23040 - 2880 = 4032 \times 5$

$20160=20160$

It is correct.

(II) P is completely divisible by A

It is correct.

(III) $P > 4320A$

$23040 > 21600$

It is correct

Sol (24 - 25):

Series I.

$a=297$ $d=401$ $c=525$ 669 833 1017 1221
 104 124 144 164 184 204
 20 20 20 20 20

Series II.

$b=300$ 318 336 354 372 390
 18 18 18 18 18

Series III.

456 472 408 552 $z = 296$
 $+16$ -64 $+144$ -256
 $+4^2$ -8^2 $+12^2$ -16^2

$a = 297, d = 401, c = 525, b = 300$ and $z = 296$

24. (e): Option (a) $a + d - c > z$

$297+401-525 > 296$

$173 < 296$

It is incorrect.

Option (b) $a + d - c < z$

$297+401-525 < 296$

$173 < 296$

it is correct

(c) $a + d = b + z + 102$

$297 + 401 = 300 + 296 + 102$

$698 = 698$

It is correct.

25. (a): (A) $d=401$ it is prime number

It is correct.

(B) $a=297$ is an odd number but $z=296$ is an even number.

It is incorrect.

$$(C) \text{ Quotient} = \frac{525}{7} = 75$$

$$75 = \frac{1}{4} \times 300$$

$$75 = 75$$

It is correct.

Sol (26 – 29): LCM of z and 5 is 15

So, z = either 3 or 15.

If z = 3

$$12 = c$$

$$\frac{3}{10} \times \frac{y}{20} = \frac{3}{10b}$$

$$by = 20$$

Possible value of b and y

$$10 = y, b = 2$$

Root of equation = 0.3, 0.5

Put in the equation given question

We get a = 5

If z = 15

$$24 = c$$

$$\frac{15}{10} \times \frac{y}{20} = \frac{3}{10b}$$

by = 4 (any multiple of y cannot give a number with unit digit zero)

Series I. a, b+5, c+8, 87, 412, 2185, 13326

5, 7, 20, 87, 412, 2185, 13326

Correct series = $5 \times 1 + 1 = 6$

$$6 \times 2 + 8 = 20$$

$$20 \times 3 + 27 = 87$$

$$87 \times 4 + 64 = 412$$

$$412 \times 5 + 125 = 2185$$

$$2185 \times 6 + 216 = 13326$$

Wrong term = 7

Series II. b+4, c+8, 36, a+50, 79, 111, 152

6 20 36 55 79 111 **155**

14 16 19 24 32 44

2 3 5 8 12

1 2 3 4

$$20 = y, b = 1$$

Root of equation = 0.3, 1

Put in the equation given question

We get a = 8.

From above we know the correct series, value of a = 5 and wrong term is 7.

So, we do not have to check this case.

26. (a): LCM of y= 10, z = 3, and c = 12.

Answer = 60

27. (e): Wrong term of series II = 152

Correct term of series I = 6

Required difference = 146

28. (e): Resultant term = $25 - 7 = 18$

29. (b): As we know z = 3

$$3 \times 1 + 1 = 4$$

$$4 \times 2 + 8 = 16$$

30. (d): Series I: Put X=24

Pattern of the series

$$\begin{array}{ccccccc} 7 & 14 & X=24 & 37 & 53 & 72 \\ 7 & 10 & & 13 & 16 & 19 \\ & 3 & & 3 & 3 & 3 \end{array}$$

So, 24 is satisfying series I.

Series II:

Pattern of the series

$$\begin{array}{ccccccc} 48 & & 25 & & 26 & & 53 & & 213 & & 1705 \\ & \times .5 + 1 & & \times 1 + 1 & & \times 2 + 1 & & \times 4 + 1 & & \times 8 + 1 \end{array}$$

So, X=24 is not satisfying series II.

Series III: Put X = 24

Pattern of the series

$$\begin{array}{ccccccc} 8 & & 15 & & X=24 & & 35 & & 48 & & 63 \\ 7 & & 9 & & & & 11 & & 13 & & 15 \end{array}$$

So, X=24 is satisfying series III.

31. (a): Pattern of the series:

$$\begin{array}{ccccccc} 18 & & 10 & & 12 & & 27 & & X=112 & & P=901 \\ & \times 0.5 + 1 & & \times 1 + 2 & & \times 2 + 3 & & \times 4 + 4 & & \times 8 + 5 \end{array}$$

$$Z + 4 = X - 27$$

$$Z = 112 - 27 - 4 = 81$$

32. (e): P = 901

(33-35):

Pattern of the series A:

$$\begin{array}{ccccccc} 24, & 31.5, & 46.5, & 69, & 99, & 136.5, & 181.5 \\ 7.5 & 15 & 22.5 & 30 & 37.5 & 45 \end{array}$$

Wrong number = 98

Pattern of the series B:

$$\begin{array}{ccccccc} 5926, & 886, & 166, & 46, & 22, & 16, & 14 \\ 5040 & 720 & 120 & 24 & 6 & 2 \\ 7! & 6! & 5! & 4! & 3! & 2! \end{array}$$

Wrong number = 18

Pattern of the series C:

$$\begin{array}{ccccccc} 11, & 18, & 44, & 107, & 231, & 446, & 788 \\ 7 & 26 & 63 & 124 & 215 & 342 \\ (2^3 - 1) & (3^3 - 1) & (4^3 - 1) & (5^3 - 1) & (6^3 - 1) & (7^3 - 1) \end{array}$$

Wrong number = 445

33. (a): X = 98, Y = 18, Z = 445

So, X > Y < Z

34. (d):

Pattern of the series:

$$\begin{array}{ccccccc} 23 & 30 & P=56 & 119 & 243 & Q=458 & 800 \\ 7 & 26 & 63 & 124 & 215 & 342 \\ (2^3 - 1) & (3^3 - 1) & (4^3 - 1) & (5^3 - 1) & (6^3 - 1) & (7^3 - 1) \end{array}$$

$$\text{Required difference} = 458 - 56 = 402$$

35. (b): P = 99, Q = 16

$$x^2 - 15x = -56$$

$$x^2 - 15x + 56 = 0$$

$$x = 8, 7$$

$$R = 8 \times 8 = 64$$

From (i)

$$Q + R = P$$

$$16 + 64 = 99$$

$$80 = 99$$

(i) does not follow

From (ii)

$$\frac{P}{3} + 15 = R - Q$$

$$\frac{99}{3} + 15 = 64 - 16$$

$$48 = 48$$

(ii) follows

From (iii)

$$\sqrt{R} + P = Q \times 7 - 5$$

$$\sqrt{64} + 99 = 16 \times 7 - 5$$

$$107 - 107$$

(iii) does follows

36. (b): Pattern of series -

$$\begin{array}{cccccc} 229, & 256, & 320, & 445, & 661, & 1004 \\ +27 & +64 & +125 & +216 & +343 & \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ 3^3 & 4^3 & 5^3 & 6^3 & 7^3 & \end{array}$$

$$X = 445$$

$$\text{For I: } \frac{445}{15} = 29.66 \text{ (I is incorrect)}$$

$$\text{For II: } (445 + 2) = 447$$

447 is divisible by 149 so it is not a prime number

(II is incorrect)

So, both I and II incorrect

37. (a): Pattern of the series (A):

$$\begin{array}{cccccc} 1332, & 612, & 492, & Y=468, & 462, & 460 \\ 720 & 120 & 24 & 6 & 2 & \\ 6! & 5! & 4! & 3! & 2! & \end{array}$$

$$\text{Fourth term of series (B)} = \frac{468-18}{5} = 90$$

First term of the series (B):

$$\begin{array}{cccccc} 954, & 234, & 114, & 90, & 84, & 82 \\ 720 & 120 & 24 & 6 & 2 & \\ 6! & 5! & 4! & 3! & 2! & \end{array}$$

38. (e): Pattern of the series

$$\begin{array}{cccccc} 49, & ?=56, & 82, & 145, & 269, & 484, & 826 \\ 7 & 26 & 63 & 124 & 215 & 342 & \\ (2^3 - 1) & (3^3 - 1) & (4^3 - 1) & (5^3 - 1) & (6^3 - 1) & (7^3 - 1) & \end{array}$$

$$?=A+6$$

$$56-6=A$$

$$50=A$$

Nearest square of A = 7

39. (d): Pattern of series A:

$$32 + (7^2 + 1) = 82$$

$$82 + (9^2 + 1) = 164$$

$$164 + (11^2 + 1) = 286$$

$$286 + (13^2 + 1) = 456$$

$$456 + (15^2 + 1) = 682$$

It follows the pattern

Pattern of series B:

$$140 + (5^2 + 1) = 166$$

$$166 + (7^2 + 1) = 216$$

$$216 + (11^2 + 1) = 338$$

$$338 + (13^2 + 1) = 508$$

$$508 + (17^2 + 1) = 798$$

It does not follow the pattern

Pattern of series C:

$$400 + (3^2 + 1) = 410$$

$$410 + 5^2 = 435$$

$$435 + (7^2 + 1) = 485$$

$$485 + (9^2 + 1) = 567$$

$$567 + (11^2 + 1) = 689$$

It does not follow the pattern

(40 -41):

Pattern of series I.

$$\begin{array}{ccccccccc} 5, & 6, & 8, & 14, & 38, & a=158, & 878, & 5918 \\ +1 & +2 & +6 & +24 & +120 & +720 & +5040 & \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ 1! & 2! & 3! & 4! & 5! & 6! & 7! & \end{array}$$

Pattern of series II.

$$\begin{array}{ccccccccc} 32, & 544, & 593, & 809, & b=834, & 898, & 907 \\ +512 & +49 & +216 & +25 & +64 & +9 & \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ 8^3 & 7^2 & 6^3 & 5^2 & 4^3 & 3^2 & \end{array}$$

Pattern of series III.

$$18 \times 2 + 4 = 40$$

$$40 \times 3 + 4 = 124$$

$$124 \times 4 + 4 = 500$$

$$500 \times 5 + 4 = 2504$$

$$2504 \times 6 + 4 = 15028$$

$$15028 \times 7 + 4 = 105200$$

$$c = 40$$

40. (a): Given, $m^2 = \sqrt{a + 98}$

$$m^2 = \sqrt{158 + 98}$$

$$m^2 = \sqrt{256}$$

$$m^2 = 16$$

$$m = 4 \text{ \& - } 4$$

For (A). When $m = 4$, then $\frac{834}{4} + 0.5 = 209$

And, 209 is an integer

When $m = -4$, then $\frac{834}{-4} + 0.5 = -208.5 + 0.5 = -208$

And, -208 is an integer

So, (A) is true

For (B). When $m = 4$, then $10 \times 4 = 40$

And $c = 40$

But when $m = -4$, then $10 \times -4 = -40$

So, we definitely can't say $10m = c$

For (C). When $m = 4$, then $4 + \frac{(834-42)}{18} = 48$

And, $c = 40$

So, resultant $> c$

When $m = -4$, then $-4 + \frac{(834-42)}{18} = 40$

And $c = 40$

So, resultant $= c$

So, (C) is true

Only (A) and (C) are true.

- 41. (e):** $n^2 - cn + (2.5a - 11)$
 $= n^2 - 40n + (2.5 \times 158 - 11)$
 $n^2 - 40n + 384 = 0$
 $n(n - 24) - 16(n - 24) = 0$
 $n = 24 \text{ \& } 16$
 So, $y = 2 \times 24 = 48$
 For (a) $(48 - 40) = 8 = 2^3$
 So, (a) is true
 For (b) Value of $\frac{834}{48} = 17.375 > 16$
 So, (b) is true
 For (c) Value of $\frac{158}{48} = 3.29$ is not an integer
 So, both (a) and (b) are true

Sol (42-44):

The pattern of the series I:

- $5 \times 1 + 1 = 6$
 $6 \times 2 + 1 = 13$
 $13 \times 3 + 1 = 40$
 $40 \times 4 + 1 = 161$
 $161 \times 5 + 1 = 806$
 So, series II will be:
 $p = 2 \times 1 + 1 = 3$
 $q = 3 \times 2 + 1 = 7$
 $r = 7 \times 3 + 1 = 22$
 $s = 22 \times 4 + 1 = 89$

- 42. (c):** $s = 89$

- 43. (d):** $2 \times 1 + 1 = 3$
 $3 \times 2 + 1 = 7$
 $7 \times 3 + 1 = 22$
 $22 \times 4 + 1 = 89$
 $89 \times 5 + 1 = 446$
 $446 \times 6 + 1 = 2677$
 So, n^{th} term = 7

- 43. (e):** Required value $= 22 - 3 \times 7 = 1$

- 44. (a):** The pattern of the series I is:

- $11 + 7^2 = 60 = P$
 $60 + 11^2 = 181$
 $181 + 13^2 = 350$
 $350 + 17^2 = 639$
 $639 + 19^2 = 1000$
 Prime number near P = 59/61
 So, 5th term of the new series $= 59 + 2 + 2 + 2 + 2 = 67$
 Or
 $61 + 2 + 2 + 2 + 2 = 69$

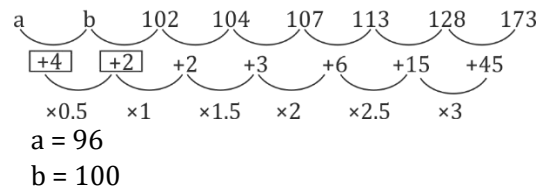
- 45. (c):** Pattern of the series II:

- $242 + 2^2 = 246$
 $246 + 3^2 = 255$
 $255 + 5^2 = 280$
 $280 + 7^2 = 329$



- $329 + 11^2 = 450$
 $450 + 13^2 = 619$
 So, wrong no. is 251.
 So, required value $= 251 \times 2 + 1 = 503$

- (47-48):**



- 47. (c):** $b = 96$

New number series
 $= 100, 100 + 2^2, 100 + 3^3, 100 + 4^4, 100 + 5^5$
 Fifth term of series $= 100 + 5^5$

- 48. (d):** $96 \times 2 = X$

$192 = X$
 4^{th} term of series given $= 104$
 Req. difference $= 192 - 104 = 88$

- 49. (a):** In the given series -

If we take $701 = X$
 So, $349 = Y$
 Similarly, $173 = X$
 So, $85 = Y$ and so on ----
 Now, (i) $701 = 2 \times 349 + 3$
 $701 = 701$
 If we take any term of given series, then statement (i) following
 (ii) $349 = 701 \times 3 + 2$
 $349 \neq 2105$
 So, statement (ii) did not follow if we take any term of the given series as 'X' & 'Y'
 (iii) $(701 - 349) = 2 \times (349 - 173)$
 $352 = 2 \times 176$
 $352 = 352$
 So, if we take any term of given series, then statement (iii) following

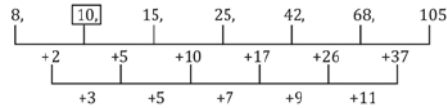
- 50. (d):** Wrong term = 3

Pattern of series -
 $1 \times 1 + 1 = 2$
 $2 \times 2 + 2 = 6$
 $6 \times 3 + 3 = 21$
 $21 \times 4 + 4 = 88$
 $88 \times 5 + 5 = 445$
 $445 \times 6 + 6 = 2676$
 So, IVth term of new formed series which starting with wrong term '3'
 $3 \times 1 + 1 = 4$
 $4 \times 2 + 2 = 10$
 $10 \times 3 + 3 = 33$

51. (e): Series I: 8, 9, 15, 25, 42, 68, 105

Wrong term = 9

Pattern of series –



So, 9 is the nearest square of '3'

52. (c): Wrong term = 1380

Pattern of series –

$$60 \times 3 = 180$$

$$180 \times 2.5 = 450$$

$$450 \times 2 = 900$$

$$900 \times 1.5 = 1350$$

$$1350 \times 1 = 1350$$

$$1350 \times 0.5 = 675$$

$$(i) \frac{1380}{30} + 1 = 47$$

47 is a prime number, so statement (i) following

$$(ii) \frac{1380}{30} + 1 = 47$$

$$\text{Sum of digits} = 4 + 7 = 11$$

And, $11 > 9$

So, statement (ii) did not follow

$$(iii) \frac{1380}{30} + 1 = 47$$

$$\frac{47}{5}, \text{ remainder is 2 not 4}$$

So, statement (iii) did not follow

Sol. (53 – 54):

We know, common factor of any prime number be 1

And HCF of prime number be 1

So, $X = 1$

$$\text{And from } z^2 - z - 6 = 0$$

We get $z = 3$ & -2

Given, $Y = \text{Smallest root of } z^2 - z - 6 = 0$

So, $Y = -2$

Now from value of X & Y we get

$$a = 19$$

$$b = 21$$

$$c = 28$$

$$d = 42$$

Now series – 20, 19, 21, 28, 42, 65

53. (c): From series we get only b & d are divisible by 3

54. (d): From series we get value of $d = 42$

55. (c): The given pattern is

$$\times 1+1 \quad \times 2+2 \quad \times 3+3 \quad \times 4+4 \quad \times 5+5 \quad \times 6+6$$

So, wrong number = 3

The new series will be

$$A = 3 \times 1 + 1 = 4$$

$$B = 4 \times 2 + 2 = 10$$

$$C = 10 \times 3 + 3 = 33$$

57. (a): Here, the pattern followed is

$$300 + (3^3 - 3) = 324$$

$$324 + (4^3 - 4) = 384$$

$$384 + (5^3 - 5) = 504$$

$$504 + (6^3 - 6) = 714 = P$$

$$714 + (7^3 - 7) = 1050$$

The next two series given follows the same pattern as above series

And according to that pattern, the value of Q and R are 534 and 614 respectively

Here, $P > Q < R$

58. (c): Here, the pattern followed is

$$2700 \times 2 = 5400$$

$$5400 \div 3 = 1800 = P$$

$$1800 \times 4 = 7200$$

$$7200 \div 5 = 1440$$

$$1440 \times 6 = 8640$$

The given next two series follow the same pattern as above series

And according to that pattern, the value of Q and R are 4200 and 4000 respectively

Here, $P < Q > R$

59. (d): Here, the pattern followed is

$$35 \times 2 = 70$$

$$70 \times 3 = 210$$

$$210 \times 4 = 840 = P$$

$$840 \times 5 = 4200$$

$$4200 \times 6 = 25200$$

The next two series follow the same pattern as above series

And according to that pattern, the value of Q and R are 840 and 960 respectively

Here, $P = Q < R$

60. (c): Here, the pattern followed is

$$(7 \times 2) + 1 = 15$$

$$(15 \times 3) + 2 = 47$$

$$(47 \times 4) + 3 = 191$$

$$(191 \times 5) + 4 = 959 = Q$$

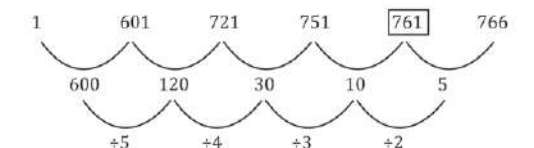
$$(959 \times 6) + 5 = 5759$$

The next two series follows the same pattern as above series

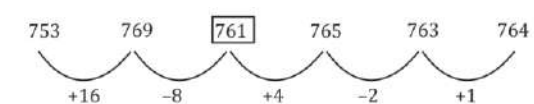
And according to that pattern, the value of P and R are 239 and 1439 respectively

Here, $P < Q < R$

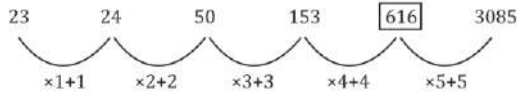
61. (d):



$$x = 761$$



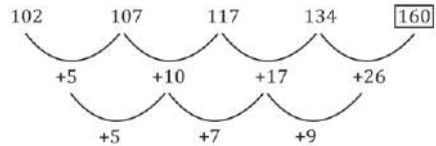
$$y = 761$$



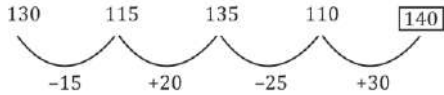
$$z = 616$$

$$\therefore x = y > z.$$

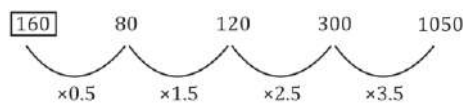
62. (b):



$$x = 160$$



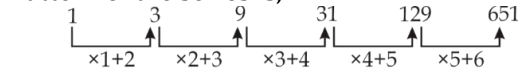
$$y = 140$$



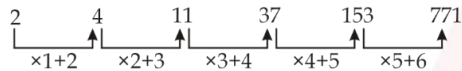
$$z = 160$$

$$\therefore x = z > y$$

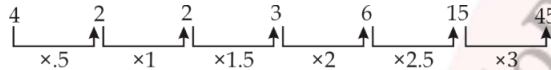
63. (d): Pattern of the series is,



Similarly,



64. (c): Pattern of the series is



So, next terms will be 45×3.5 , $45 \times 3.5 \times 4$, $45 \times 3.5 \times 4 \times 4.5$

Do not calculate exact values. Just calculate approximate values because with increase in values next term will be far away from 2835

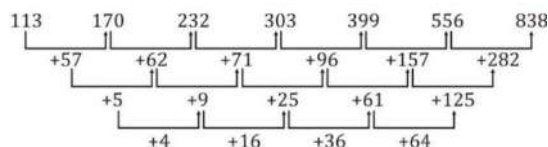
So, 45×3.5 is approx. 150

And, 150×4 is approx. 600

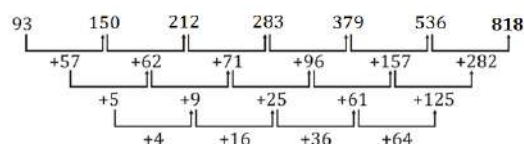
600×4.5 is approx. 2700

So, multiplies of 4.5 is nth term which is 10th term.

65. (d):



Second series



Sol. (67-69):

Series I: Pattern of series -

$$48 \times 1.25 = 60$$

$$60 \times 2.5 = 150$$

$$150 \times 5 = 750$$

$$750 \times 10 = 7500$$

$$7500 \times 20 = 150000$$

Series II. Pattern of series -

$$1500 \times \frac{4}{5} = 1200$$

$$1200 \times \frac{3}{5} = 720$$

$$720 \times \frac{2}{5} = 288$$

$$288 \times \frac{1}{5} = 57.6$$

$$57.6 \times \frac{0}{5} = 0$$

Series III. Pattern of series -

$$15 + (3^2 - 1) = 23$$

$$23 + (4^3 - 1) = 86$$

$$86 + (5^2 - 1) = 110$$

$$110 + (6^3 - 1) = 325$$

$$325 + (7^2 - 1) = 373$$

67. (c): P = 7500

$$Q = 1200$$

$$R = 373$$

$$\text{For (a) } 7500 > 1200 > 2 \times 373$$

So, (a) not correct relation among P, Q and R

$$\text{For (b) } \frac{7500}{5} < 1800 > 373$$

So, (b) not correct relation among P, Q and R

$$\text{For (c) } 7500 = (7200 + 300) > 7460$$

So, (c) correct relation among P, Q and R

$$\text{For (d) } 7500 > 7200 > 5595$$

So, not correct relation among P, Q and R

68. (b): Second term of series IV = $(5625 - 5 \times 1200 + 372) = -3$

Let first term of series IV be x

$$x + (3^2 - 1) = -3$$

$$x + 8 = -3$$

$$x = -11$$

So, series IV will be

$$-3 + (4^3 - 1) = 60$$

$$60 + (5^2 - 1) = 84$$

$$\text{Required difference} = 84 - (-11) = 95$$

69. (b): X = 5625

$$Y = 1000$$

$$Z = 372$$

Pattern of series:

$$\begin{array}{ccccccc} 6 & A=8 & 14, & 38, & 158, & 878, & B=5918 \\ +2 & +6 & +24 & +120 & +720 & +5040 & \\ 2! & 3! & 4! & 5! & 6! & 7! & \end{array}$$

$$\text{For (i) } 5625 + 372 + 10 = 5718 + 289 - 200$$

$$6007 = 6007$$

So, (i) true

$$\text{For (ii) } 1000 = 8 \times 125$$

$$1000 = 1000$$

So, (ii) true

$$\text{For (iii) } \sqrt[3]{8} + 372 = \frac{1000}{2.5}$$

$$374 \neq 400$$

So, (iii) not true

Sol. (70-71):**From series B:** Pattern of series –

$$374 - 327 = 47$$

$$(441 + 7) - 374 = 448 - 374 = 74$$

$$495 - 448 = 47$$

So, the pattern of series B: +47, +74, +47, +74

$$x + 36 + 47 = 132$$

$$x = 49$$

$$y = 132 + 74 = 206$$

$$206 + 47 = b^2 - 3$$

$$b^2 = 256$$

$$b = 16$$

From series A: $z = 2 \times (441 + 7) + (82)$

$$z = 896 + 82$$

$$z = 978$$

$$\text{third term of series A} = (9 \times 206 - 35) = 1819$$

Form first, second and third term of series A

$$17 + 31^2 = 978$$

$$978 + 29^2 = 1819$$

$$(2304 + 44) - 1819 = 529$$

So, pattern of series –

Addition of square of consecutive prime numbers

$$(2304 + 44) + 19^2 = 2500 + a$$

$$2348 + 361 = 2500 + a$$

$$a = 209$$

$$70. (d): P = (209 + 206) = 415$$

$$\text{For I. } 415 = 8 \times 49 + (16 + 7)$$

$$415 = 392 + 23$$

$$415 = 415$$

So, I true

$$\text{For II. } 415 = \frac{49}{2} \times 16 + 23$$

$$415 = 392 + 23$$

$$415 = 415$$

So, II true

$$\text{For III. } 415 \times 2 = 978 - 148$$

$$830 = 830$$

So, III true

All I, II and III true

$$71. (e): \text{Given, } 2^p \times 7^q = bx$$

$$2^p \times 7^q = 16 \times 49$$

$$2^p \times 7^q = 2^4 \times 7^2$$

$$\text{So, } p = 4$$

$$\text{And } q = 2$$

$$\left(\frac{z}{2} - 2a\right)^{p-q} = \left(\frac{978}{2} - 2 \times 209\right)^{4-2}$$

$$= (489 - 418)^2$$

$$= 71 \times 71$$

$$= 5041$$

$$72. (e): \text{Pattern of series I:}$$

$$76 + 320 = 396 = X$$

$$396 + 160 = 556$$

$$556 + 80 = 636$$

$$636 + 40 = 676$$

$$676 + 20 = 696$$

$$Y = 696 + 10 = 706$$

Pattern of series II.

$$\begin{array}{ccccccccccc} A = 136, & 148, & 177, & 223, & 286, & B = 366, & 463 \\ & +12 & +29 & +46 & +63 & +80 & +97 \\ & & +17 & +17 & +17 & +17 & +17 \end{array}$$

$$\text{For P. } 706 = 5 \times 136 + 26$$

$$706 = 380 + 26$$

$$706 = 706$$

is correct

$$\text{For Q. } X - B = 396 - 366 = 30$$

B is correct

$$\text{For R. } 2 \times 136 + 706 < 2 \times 366 + (396 - 144)$$

$$978 < 984$$

R is incorrect

So, only R is incorrect

Sol. (73- 74):**Pattern of series I:**

$$8 + (1^2 + 1) = 10$$

$$10 + (2^3 + 2) = 20$$

$$20 + (3^2 + 1) = 30$$

$$30 + (4^3 + 2) = 96$$

$$96 + (5^2 + 1) = 122$$

$$122 + (6^3 + 2) = 340 = x$$

$$340 + (7^2 + 1) = 390$$

Pattern of series II.

$$16 \times \frac{1}{2} + 2 = 10$$

$$10 \times \frac{2}{3} + 3 = 18$$

$$18 \times \frac{1}{2} + 2 = 11$$

$$11 \times \frac{3}{2} + 3 = 19.5$$

$$19.5 \times \frac{1}{2} + 2 = 11.75 = y$$

$$11.75 \times \frac{3}{2} + 3 = 20.625$$

Pattern of series III.

$$26 + 2 = 28$$

$$28 \times 4 = 112$$

$$112 + 6 = 118$$

$$118 \times 8 = 944$$

$$944 + 10 = 954 = z$$

$$954 \times 12 = 11448$$

$$73. (d): a^2 = \sqrt{(340 + 2) + 954}$$

$$a^2 = \sqrt{1296}$$

$$a^2 = 36$$

$$a = 6 \text{ and } -6$$

For (i) When we take, $a = 6$

$$\text{Then, } 2 \times 6 = (11.75 + 0.25)$$

$$12 = 12$$

When we take, $a = -6$

$$2 \times -6 < (11.75 + 0.25)$$

$$-12 < 12$$

So, (i) is true

For (ii) When we take, $a = 6$

$$2^6 > 11.75$$

$$64 > 11.75$$

When we take, $a = -6$

$$2^{-6} < 11.75$$

$$\frac{1}{64} < 11.75$$

So, (ii) is not true

For (iii) When we take, $a = 6$

$$340 + 6 \times 60 < 954$$

$$700 < 954$$

When we take, $a = -6$

$$340 + (-6 \times 60) < 954$$

$$-20 < 954$$

So, (iii) is true

74. (e): $p^2 - 4 \times 11.75p + \frac{3}{2} \times 340$

$$p^2 - 47p + 510 = 0$$

$$p^2 - 30p + 17p + 510 = 0$$

$$p(p-30) - 17(p-30) = 0$$

$$p = 30 \text{ and } 17$$

$$\text{So, } b = 3 \times (30 - 17) = 39$$

For (a) $2(340 - \frac{954}{3}) - 5 = 39$

$$44 - 5 = 39$$

So, (a) is true

For (b) Value of $\frac{954}{39} \approx 24.46 > 18$

So, (b) is true

For (c) Value of $4 \times 11.75 \times 39 = 1833$

So, all (a), (b) and (c) are true

Sol. (75 - 76):

Pattern of the series:

z,	1730,	2594,	3890,	7130,	x = 18470	y = 69500
+1728	+864	+1296	+3240	+11340	+51030	
$\times 0.5$	$\times 1.5$	$\times 2.5$	$\times 3.5$	$\times 4.5$		

75. (e): New series will be

$$69500 \div 5 = 13900$$

$$69500 \div 10 = 6950$$

$$69500 \div 20 = 3475$$

$$69500 \div 40 = 1737.5$$

$$5^{\text{th}} \text{ term} = 1737.5$$

(a) 1737.5 is not an integer number, so (a) is not true

(b) $1737.5 < 1800.5$, so (b) is true

(c) 1737.5 is not a whole number, so (c) is not true

(d) $1737.5 > (12)^3 = 1728$, so (d) is true

So, (b) and (d) both true

76. (e): $A = (\frac{18470}{50} - 45.4) = (369.4 - 45.4) = 324$

So, value (approx.) of $(y \div A) = (69500 \div 324)$
 $= 214.506 \approx 214.5$

I. $214.5 < 216$, so I is true

II. $214.5 > 196$, so II is true

$$\text{III. } (214.5)^2 = 46010.25$$

46010.25 is not more than y, so III is not true

Sol. (77 - 79):

Pattern in the series I.

119,	125,	135,	153,	X=183,	229,	295
6	10	18	30	46	66	
	4	8	12	16	20	

Pattern in the series II.

4,	28,	168,	840,	3360,	Y=10080,	20160
$\times 7$	$\times 6$	$\times 5$	$\times 4$	$\times 3$	$\times 2$	

Pattern in the series III.

105,	111,	Z=99,	119,	89,	131,	75
$+(2 \times 3),$	$-(3 \times 4),$	$+(4 \times 5),$	$-(5 \times 6),$	$+(6 \times 7),$	$-(7 \times 8)$	

77. (c): $X = 183, Y = 10080, Z = 99$

$$\text{So, } Y > X > Z$$

78. (b): From (i)

$$X = 183 \text{ and } Z = 99$$

$$\text{H.C.F. of } X \text{ \& } Z = 3$$

$$\sqrt[3]{27} = 3$$

So, (i) is true.

From (ii)

$$= \frac{Y}{Z-9} + 32$$

$$= \frac{10080}{99-9} + 32$$

$$= 144 \text{ is a perfect square of } 12.$$

So, (ii) is true.

From (iii)

$$\text{Req. difference} = \frac{10080-183}{4} = \frac{9897}{4} = 2474.25$$

So, (iii) is not true.

79. (c): Pattern of the series IV

$$183, 189, 177, 197, 167, 209, 153$$

$$+(2 \times 3), \quad -(3 \times 4), \quad +(4 \times 5), \quad -(5 \times 6), \quad +(6 \times 7), \quad -(7 \times 8)$$

$$\text{Fifth term} = 167$$

80. (b):

Pattern of series I.

5,	6,	8,	14,	38,	a=158,	878,	5918
+1	+2	+6	+24	+120	+720	+5040	
↑	↑	↑	↑	↑	↑	↑	
1!	2!	3!	4!	5!	6!	7!	

Pattern of series II.

32,	544,	593,	809,	b=834,	898,	907
+512	+49	+216	+25	+64	+9	
↑	↑	↑	↑	↑	↑	
8 ³	7 ²	6 ³	5 ²	4 ³	3 ²	

Pattern of series III.

$$18 \times 2 + 4 = 40$$

$$40 \times 3 + 4 = 124$$

$$124 \times 4 + 4 = 500$$

$$500 \times 5 + 4 = 2504$$

$$2504 \times 6 + 4 = 15028$$

$$15028 \times 7 + 4 = 105200$$

$$c = 40$$

$$P = (2 \times 40 + 158 + 834) = 1072$$

$$\text{Nearest square of } 1072 = 1089$$

$$1089 \text{ is square of } 33.$$

So, 33 will be correct

